

EXHALE: User Manual

Scope. This manual describes how to set up, run, and analyze models with EXHALE, the in-house extension of the ATES (ATmospheric EScape) code to trace metals, metal-line cooling, a runtime opacity dispatcher, the Roche tidal potential, and excited-hydrogen / Ly α physics. It documents every input parameter (with a brief physical meaning), every output file (column by column, with units), the analysis tools, and a set of worked examples. The detailed physics derivations live in the companion notes (Update_EXHALE_early_phase.tex, transmission_spectrum.tex, Update_EXHALE.tex, cooling_formulas.tex — the complete reference for every analytic cooling coefficient — EXHALE_BC_and_IC.tex — the boundary- and initial-condition reference — and wind_ae_solver.tex — the Wind-AE steady-wind solver used by IC mode: windae); this document is the practical user guide.

Contents

1. Introduction	2
1.1 What the code does	2
1.2 The lower boundary, radii, and normalization	3
1.3 The gravitational potential	3
1.4 Workflow	4
2. Installation and running	4
2.1 Requirements	4
2.2 Build	5
2.3 Running	5
2.4 Convergence control	5
2.5 Base grid resolution (Base_grid [dr,cells])	6
2.6 The base ghost temperature (Base_ghost_temperature)	7
2.7 The steady-state Newton solver (Solver: Newton)	8
2.8 Environment-variable hooks (diagnostics and expert runs)	9
2.9 Metal cooling on/off comparison	9
2.10 Wind-AE warm-start initial conditions	11
3. Input files	12
3.1 input.inp — the main parameter file	12
3.2 metals.inp — trace-metal abundances	28
3.3 opacity.inp — opacity model (optional)	28
3.4 Excited hydrogen and the Ly α pumping field	28
3.5 Spectral coverage for metal photoionization	29
3.6 Diffusive separation of helium and metals	30
3.7 Lower-atmosphere connection (Tier 1–3)	31
3.8 Obtaining VULCAN and FastChem	33

4. Output files	34
4.1 Hydro_ioniz.txt / Hydro_ioniz_adv.txt	35
4.2 Ion_species.txt / Ion_species_adv.txt	35
4.3 Cooling_breakdown.txt	36
4.4 Heating_breakdown.txt	36
4.5 Lyman_Werner.txt	36
4.6 Excited_H.txt	36
4.7 EXHALE_setup.out and the mass-loss rate	37
5. Analysis tools	37
5.1 EXHALE_plots.py — live profile viewer	37
5.2 EXHALE_transit.py — transmission spectra	37
5.3 exhale_io.py — named, physical-unit loaders	40
5.4 exhale_to_lart.py — LaRT Ly α input from an EXHALE run	40
6. Worked examples	42
6.1 A first run: a generic hot Jupiter	42
6.2 Turning physics on and off	43
6.3 Ready-made example configurations (examples/)	43
7. Comparison studies	48
7.1 Effect of metal-line cooling (metals on vs. off)	48
7.2 Spherical vs. Roche (RLOF) and the mass-loss rate	48
7.3 Transmission spectra	48
8. Atomic and molecular data	50

1. Introduction

1.1 What the code does

EXHALE solves the time-dependent, spherically-symmetric (1-D) radiation–hydrodynamics of an irradiated planetary upper atmosphere and the hydrodynamic wind it drives (“atmospheric escape”). It integrates the Euler equations for mass, momentum, and energy of a single fluid,

$$\partial_t \rho + \frac{1}{r^2} \partial_r (r^2 \rho v) = 0, \quad \partial_t (\rho v) + \frac{1}{r^2} \partial_r (r^2 \rho v^2) + \partial_r p = -\rho \partial_r \Phi,$$

$$\partial_t E + \frac{1}{r^2} \partial_r (r^2 v (E + p)) = -\rho v \partial_r \Phi + \mathcal{H} - \mathcal{C},$$

to a steady state, where Φ is the gravitational (optionally Roche) potential (§1.3), $\mathcal{H}$ is the XUV photoionization heating and $\mathcal{C}$ the radiative cooling. Coupled to the hydrodynamics, at every step it solves the *ionization equilibrium* of H, He, and the trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) with a Newton (MINPACK) solver, including photoionization,

collisional ionization, radiative + dielectronic recombination, and charge exchange. The photoelectron energy budget carries Shull & van Steenberg (1985) secondary ionization (a fast photoelectron drives further H I / He I ionizations instead of thermalizing fully), the He II $\rightarrow$ He I recombination radiation is coupled to H I ionization on the spot (Draine 2011 y/z), and the He $\leftrightarrow$ H charge-exchange pair (Koskinen 2013) acts in every helium-bearing ionization system. When the metastable He 2^3S triplet is included the code adds its explicit collisional-excitation cooling (the 10830 Å channel) and the He $2^3\text{S} + \text{H} / \text{H}_2$ Penning ionization that removes the metastable. The converged run yields the radial profiles of density, velocity, temperature, and the ionization state, plus the steady-state *mass-loss rate* $\dot{M}$. A post-processing pass adds advection to the ionization balance, and the Python tool `EXHALE_transit.py` turns a converged model into transit *transmission spectra* (He 10830, Ly α , H α /H β , and the metal resonance lines Mg II, Ca II, Na I).

1.2 The lower boundary, radii, and normalization

The computational domain starts at a **lower boundary** placed at the 1 μbar pressure level, which defines $r = 1$ and the base radius R_p (the input `Planet radius`). The base number density n_0 and temperature T_0 (the equilibrium temperature) are set there. Internally the code is non-dimensional: lengths in R_p , velocities in $v_0 = \sqrt{k_B T_0 / \mu}$, densities in n_0 , etc. *All output files are written back in cgs* (see §4. for the exact normalization of each column). The key dimensionless number is the *Jeans / escape parameter* $b_0 = GM_p \mu / (k_B T_0 R_p)$, the depth of the gravitational well in units of the thermal energy; small b_0 means an easily-escaping (or tidally unbound) atmosphere.

1.3 The gravitational potential

The potential Φ in the momentum and energy equations is selected by `Domain mode` (Table 3). The code writes it in the dimensionless units introduced above (lengths in R_p , potential in $v_0^2 = k_B T_0 / \mu$), so it is fixed by just three numbers: the Jeans parameter $b_0 = GM_p \mu / (k_B T_0 R_p)$, the stellar-to-planet mass ratio $q = M_\star / M_p$, and the orbital separation $\tilde{a} = a / R_p$.

Spherical mode (`Domain mode: Spherical`). Only the planet’s own gravity is kept, so the domain can extend far past the Hill radius:

$$\phi(r) = -\frac{b_0}{r}.$$

Roche mode (default). The full co-rotating Roche potential, evaluated along the star–planet (substellar) line $x = r$, adds the stellar tidal attraction and the centrifugal term of the synchronously-rotating frame:

$$\phi(r) = -\frac{b_0}{r} - \frac{b_0 q}{\tilde{a} - r} - \frac{b_0(1+q)}{2\tilde{a}^3} \left(\frac{q\tilde{a}}{1+q} - r \right)^2,$$

where the second term is the star’s gravity at distance $\tilde{a} - r$ and the third is the centrifugal potential about the center of mass at $x_{\text{cm}} = q\tilde{a} / (1+q)$ with orbital frequency $\Omega^2 = G(M_p + M_\star) / a^3$ (synchronous rotation). The effective gravity used by the hydrodynamics is $g(r) = -d\phi/dr$, with

$$\frac{d\phi}{dr} = \frac{b_0}{r^2} - \frac{b_0 q}{(\tilde{a} - r)^2} + \frac{b_0(1+q)}{\tilde{a}^3} \left(\frac{q\tilde{a}}{1+q} - r \right).$$

The inner Lagrange point L_1 is the root of $d\phi/dr = 0$, where planetary gravity, the stellar tide, and the centrifugal force cancel; in Roche mode the outer boundary is placed at this L_1 radius by default, beyond which gas is no longer bound to the planet.

Roche mode with an extended outer boundary (Outer radius in Roche mode). By default Roche mode truncates at L_1 . Supplying an explicit `Outer radius [R_p]: > 1 without Domain mode: Spherical` now keeps the full tidal Roche potential above *and* pushes the outer boundary out to the requested radius (e.g. $10R_p$), reproducing the extended-domain treatment of Yan et al. (2022) / Huang et al. (2023), whose escape models retain the tidal force but integrate well past L_1 . This differs from `Domain mode: Spherical`, which also extends the domain but *drops* the tidal and centrifugal terms (pure $-b_0/r$). Two caveats apply. First, beyond L_1 the Roche potential is past its maximum, so the net force is outward and the 1-D radial flow there is a spherical approximation to what is physically an L_1 funnel — the same approximation Yan/Huang make. Second, the cold hydrostatic IC is invalid past L_1 ; use `IC mode: auto` (or `transonic`) so the solver seeds a transonic wind through the interior sonic point. The tidal singularity of the $-b_0 q/(\tilde{a}-r)$ term sits at the star ($r = \tilde{a} \gg 10$), so a $10R_p$ domain is numerically safe.

Full 3-D form (transmission post-processing). The 3-D Roche-equipotential reconstruction in `EXHALE_transit.py` / `roche_recon.py` (§5.2) needs the potential off-axis as well. With the substellar direction $\hat{x}$ pointing toward the star and (y, z) the sky plane,

$$\phi(x, y, z) = -\frac{b_0}{\sqrt{x^2 + y^2 + z^2}} - \frac{b_0 q}{\sqrt{(\tilde{a}-x)^2 + y^2 + z^2}} - \frac{b_0(1+q)}{2\tilde{a}^3} \left[(x_{\text{cm}} - x)^2 + y^2 \right],$$

which reduces to the substellar $\phi(r)$ above on the $+x$ axis. The centrifugal term involves only the rotation-plane coordinates (x, y) , not z , so surfaces of constant ϕ are triaxial (most extended toward L_1 , least along the spin axis z).

1.4 Workflow

A model run consists of:

1. Prepare `input.inp` (planet/star parameters, numerics, physics switches) — by hand or via the GUI; optionally `metals.inp` (trace metal abundances) and `opacity.inp` (opacity model).
2. Build and run `./EXHALE.x`; it reads the input files from the current directory and writes profiles to `output/`.
3. Inspect with `EXHALE_plots.py` (live profiles) and the example notebook; compute transmission spectra with `EXHALE_transit.py`.

2. Installation and running

2.1 Requirements

A Fortran compiler (gfortran by default; Intel ifort/ixf supported), make, and — for the GUI and analysis — Python 3 with tkinter, numpy, scipy, matplotlib, and astropy (the last is used by `EXHALE_transit.py` for `astropy.convolution`).

2.2 Build

EXHALE uses an incremental Makefile; module dependencies are auto-generated, so only edited sources and their dependents recompile.

```
make                # build ./EXHALE.x with gfortran
make FC=ifort       # Intel ifort (or FC=ifx for the LLVM compiler)
make wind_ae_ic     # standalone Wind-AE IC generator -> ./wind_ae_ic.x
make check          # byte-identical regression against the stored goldens
make clean          # remove .o/.mod (keeps EXHALE.x)
make distclean      # wipe build/ and EXHALE.x
```

`make check` rebuilds, re-runs each regression case single-threaded (`OMP_NUM_THREADS=1`, so the result is deterministic) and *bitwise*-compares output/Hydro_ioniz.txt and output/Ion_species.txt against the stored reference outputs. The standing rule it encodes: a refactor with no intended physics change must be byte-identical, and a new physics option must default to off. The harness and its reference outputs live in backup/regression/; the target reports that it found no harness, without failing, in a checkout that does not carry it. `make wind_ae_ic` builds the out-of-process IC generator described in §2.10.

2.3 Running

The driver script `run_EXHALE.sh` (1) launches a Python/Tk GUI that writes `input.inp`, (2) builds, and (3) runs `./EXHALE.x`:

```
chmod +x run_EXHALE.sh
./run_EXHALE.sh          # gfortran (GUI -> build -> run)
./run_EXHALE.sh --ifort  # Intel ifort (--ifx for ifx)
```

You may equally skip the GUI: place a hand-written `input.inp` (and optional `metals.inp`/`opacity.inp`) in a run directory, ensure an `output/` subdirectory exists, and run `./EXHALE.x` there. Each modeled planet typically lives in its own subdirectory holding its `input.inp`, its `EXHALE.x`, and its `output*/` folders. Progress (iteration counter, the momentum-variation `du`, and the time-derivative norm `dtu`) is printed to the terminal; the run stops when `du` or `dtu` falls below its convergence threshold.

2.4 Convergence control

The time loop stops when *any* of the following criteria is met (defaults in `src/modules/init/parameters.f90`; most are runtime-settable, Table 3):

Parameter	Default	Stop condition
<code>du_th</code>	1×10^{-3}	momentum non-uniformity $du < du_th$
<code>dtu_th</code>	1×10^{-8}	time-derivative ∞ -norm $dtu < dtu_th$
<code>stall_tol</code> , <code>N_stall</code>	10^{-6} , 2000	du on a plateau for <code>N_stall</code> steps
<code>count_max</code> (Max steps:)	10^6	hard iteration cap

The stall and count_max backstops guarantee termination even when the strict du/dtu thresholds are physically unreachable for a given setup (e.g. a transonic wind whose du floors at $\sim 1\%$). Each iteration prints count, du , dtu (and the windowed flux-level change when Level_tol: is set), and the run reports which criterion stopped it. The optional Force_start forces the first 1000 iterations regardless (useful when starting near a fixed point).

Two of these triggers are *armed* before they can fire. A generated initial condition (cold hydrostatic, transonic, warm-seed, Wind-AE) is an analytic profile the marching loop has not touched, so its du measures the smoothness of the formula that built it, not relaxation of the wind, and it can start orders of magnitude below every threshold. The $du < du_{th}$ stop and the Newton hand-off (together with the secondary-ionization flip that shares its du comparison) therefore require du to have been seen at or above their own threshold first, so only a *descending* crossing stops the run; both arm immediately when the run started from Load_IC? True, whose state was relaxed by the run that wrote it. Arming is reported in the log. The PLM $\rightarrow$ WENO3 switch is deliberately not guarded. The rationale and history of these settings are in Update_EXHALE_solver.

Caveat: du can stop a run prematurely. du measures only the *spatial spread* of $\rho v r^2$ in the wind window; it is blind to a uniform drift of the flux level, and on strongly irradiated planets it *oscillates*, dipping below 10^{-3} transiently long before the wind settles. A WASP-121b reference stopped this way was low by ~ 0.08 dex in $\dot{M}$ with the heating/cooling grossly out of balance (energy residual ~ 30). For production answers use the residual-based stop (Resid_tol:) or the Newton finish (Solver: Newton, §2.7); details in steady_solver_memo.

Staged secondary ionization. With Secondary_ionization: True (the default) the Shull & van Steenberg (1985) coupling is *not* applied from step 0. From a cold initial condition it would deepen the base cooling valley and amplify the base acoustic mode into a NaN runaway on high-gravity cases. Instead the wind first relaxes without the coupling; the moment that first relaxation converges (or stalls) the coupling switches on (the terminal prints secondary ionization activated at step N), the convergence logicals and flux-level history are reset, and stops are held for N_stall steps while the base adjustment wave forms, after which the run re-converges with the full physics. Expect roughly double the step count on high-gravity cases. Secondary_ionization: False disables the coupling entirely; Immediate applies it from step 0 (for A/B tests only — it reproduces the cold-start crash on high-gravity cases).

2.5 Base grid resolution (Base_grid [dr,cells])

The Mixed grid stacks cells uniform cells of size dr (in R_p) on the lower boundary and fills the rest of the domain with N -cells geometrically stretched cells; the total cell count N is set by the optional key Grid_cells: (default 500). The two numbers share one input line because they are not independent: their product is the radial extent of the uniform region ($0.01 R_p$ with the default 2.0×10^{-4} 50).

The quantity that matters physically is how many uniform cells span the base density scale height, H/dr with $H = kT/(\mu g)$. Where H/dr is of order a few cells the discretization supports a *stationary* $2dr$ entropy (contact) mode that nothing in the scheme removes: HLLC resolves a contact of zero speed exactly and so adds no dissipation to it, the gravity source is cell-local, the WENO3 pressure gradient sees only interface pressures, and there is no conduction or physical viscosity unless they are switched on. The measured dependence is an alternating $\ln \rho$ amplitude of $\sim 10^{-2}$

for $H/dr < 5$ and $\sim 10^{-4}$ for $H/dr > 100$. High-gravity planets have the least margin — at T_{eq} on the default grid HD 189733 b spans only ~ 27 cells per H , against ~ 60 – 100 for HD 209458 b, WASP-52 b and WASP-121 b — and a cool, shielded base shrinks H further. The value in effect is echoed in EXHALE_setup.out as `Base scale-height resolution: H(T_eq)/dr`, with a warning below 10 cells.

Refine at fixed extent, i.e. divide dr and multiply `cells` by the same factor: `Base grid [dr, cells]: 5.0e-5 200` is the $4\times$ refinement. Two costs follow. The CFL time step is set by the smallest cell, so a $4\times$ finer base means $\sim 4\times$ more steps for the same physical time; and at fixed N the stretched region loses those cells and coarsens (for a $4.43 R_p$ domain, the cell size at $1.5 R_p$ grows from 6.1×10^{-3} to $1.3 \times 10^{-2} R_p$). Raising `Grid cells`: alongside the refinement keeps the upper stretch fixed instead — the pairs `1.0e-4 100/500`, `5.0e-5 200/658`, `2.5e-5 400/916` and `1.25e-5 800/1375` hold the stretch ratio at its default-grid value — at the price of proportionally more work in every sweep. In practice the $2\times$ step (`1.0e-4 100`) is the one to try first: it is where the mode goes, and $4\times$ is not reachable to convergence on HD 189733 b within the compile-time step cap because its time step is $4\times$ smaller. That $2\times$ setting is what HD189733b/input.inp carries; the other three production planets run on the default. The full investigation is docs/hd189_base_checkerboard.md.

2.6 The base ghost temperature (Base ghost temperature)

The lower ghost cells always pin the density to its base value ρ_{bc} (the mass reservoir). Their *temperature* is a separate choice, and the legacy one is to pin it as well: the ghost pressure is fixed at $n_{\text{tot},\text{bc}} + \delta p_{\text{bc}}$, which at the pinned density is $T_{\text{ghost}} = T_0$, the equilibrium temperature. Nothing in the model supports that value — the radiative equilibrium of the lower atmosphere is outside the domain — and it becomes actively harmful when the first interior cell settles far below T_0 : with the CHIANTI metal-line cooling, cell 1 on HD 189733 b sits at 494 K against $T_0 = 1183$ K, so a factor-2.4 contact discontinuity is held permanently on the boundary, together with a density inversion ($\rho_1/\rho_{\text{ghost}} = 2.3$).

`Base ghost temperature: continuous` replaces the pin with $dT/dr = 0$ at the base face: the ghost keeps the base composition ($n_{\text{tot},\text{bc}}$ nuclei and δp_{bc} electrons at ρ_{bc} , exactly the particle count the isothermal pin uses) but carries the temperature of the first interior cell,

$$p_{\text{ghost}} = (n_{\text{tot},\text{bc}} + \delta p_{\text{bc}}) T_1, \quad T_1 = p_1 / (n_{\text{tot}} + n_e)_1, \quad (1)$$

in the code's T_0 units. The default is isothermal (the legacy pin), so runs without the key are unchanged. The key is ignored when `Hydrostatic base: True` is set, which sets the same ghost pressure from the interior gradient instead; with `Base BC: pressure` the microbar target is imposed at T_0 when n_0 is derived, so under a continuous- T ghost only the base *density* stays anchored.

Measured on HD 189733 b (production configuration, restarted from the converged isothermal-ghost state and Newton-finished): the boundary jump disappears ($T_{\text{ghost}} = 540$ K against $T_1 = 540$ K, $\rho_1/\rho_{\text{ghost}} = 0.95$), the alternating $\ln \rho$ amplitude over cells 1–12 falls from 2.6×10^{-2} to 9.2×10^{-3} , the steady residual from 5.4×10^{-4} to 2.8×10^{-4} , and beyond cell 7 the alternation drops by one to two orders of magnitude; $\log_{10} \dot{M}$ moves by $+0.01$ dex (9.04 to 9.05). What remains in that A/B is a stronger 3-cell disturbance in cells 2–4; it is much reduced in the production state that HD189733b/ has carried since, where the key is combined with the ionization-root validation ($T_{\text{ghost}}/T_1 = 529.4/529.6$ K, cells 1–12

amplitude 1.9×10^{-3}). With the later cooling and $H(n=2)$ corrections that folder now converges at a 551 K base and $\log_{10} \dot{M} = 9.14$, the value paper/ carries. Details and the full window sweep: docs/hd189_base_checkerboard.md.

2.7 The steady-state Newton solver (Solver: Newton)

The finite-volume steady state can be solved directly — $F(Y) = 0$ for the hydro state, with the ionization equilibrium eliminated locally inside the residual — by a preconditioned Jacobian-free Newton–Krylov method (banded LAPACK preconditioner, frozen WENO weights, diagonal scaling, non-monotone line search; the smooth base valve Valve eps: is required and auto-adopted). Each unknown is scaled by its own cell’s state — ρ , $\rho(|v| + c_s)$, E — so that no cell sets another’s scale; the momentum scale uses the fastest signal speed because ρv itself vanishes wherever the flow reverses. This reaches residual levels unattainable by marching at a small fraction of the cost: WASP-121b converges in 8 Newton iterations (~ 200 marching-steps of cost, vs. $> 35,000$ insufficient marching steps), HD189733b in 33 s where marching spent > 3 h without converging.

Two ways to use it:

- **Production:** add Solver: Newton to input.inp. The run marches (including the two-stage PLM→WENO3 when Reconstruction scheme: PLM+WENO3 is set) until the flux metric du (the radial spread of $\rho v r^2$) drops below the hand-off threshold newton_du_switch (default 10^{-2} ; the du -based convergence stops are suspended), then the Newton solve finishes to Resid tol (default 10^{-3}), and the standard outputs and post-processing follow. The hand-off point is adjustable through an optional third token on the key, Solver: Newton <du_switch> (e.g. Solver: Newton 5e-3 hands off later); a bare Solver: Newton keeps the 10^{-2} default.
- **From an existing state** (recommended for re-converging old runs): put the state in output/*_IC.txt, set Load IC? True, Valve eps: 1.0e-4, Resid tol: 1.0e-3, and run with the environment hook

```
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x
```

which solves from the loaded state, writes output/, and stops.

The warm start matters: Newton has a finite basin, so the hand-off must wait until the marching warm-up has flattened the wind. The default $du < 10^{-2}$ is calibrated to land inside the basin — handing off earlier (a larger newton_du_switch) risks starting Newton outside it and stalling, whereas a cold-IC start (no warm-up at all) stalls outright. Earlier revisions keyed the hand-off on the residual itself ($\|R\| < 5 \times 10^{-2}$); the du -based key is equivalent in practice and consistent with the flux-based convergence decision (§2.4), while the residual remains the *target* of the JFNK finish. Convergence is linear (frozen-weight refresh and the chemistry noise floor). Planets with a breathing base (HD189733b) used to stop at a practical floor $\|R\| \sim 3 \times 10^{-4}$; the 2026-08-11 line-search and base-energy fixes removed it, and a warm re-convergence of HD189733b now reaches $\|R\| = 1.2 \times 10^{-5}$ in 11 outer iterations (re-measured 2026-08-15). The Newton finish is also *safe to fail*: the solver tracks the best iterate seen, aborts once the line search has failed to find any acceptable step in 12 consecutive outer iterations, and on failure the run prints JFNK failed, keeps the best iterate, and falls back to plain time-marching with the du -based stops re-armed — an unconverged Newton state is never accepted as the final answer. The associated diagnostic hooks (EXHALE_RESIDUAL,

EXHALE_JAC_TEST, EXHALE_NEWTON_TEST, ...) are environment variables, collected in §2.8 (Table 1). Full development record and validation: `steady_solver_memo.pdf`. Figure 1 shows the production pipeline end-to-end on WASP-121b (Roche): the monitored residual during the marching warm-up (left; note that the du -based stop would have fired long before the residual is actually small) and the JFNK finish (right).

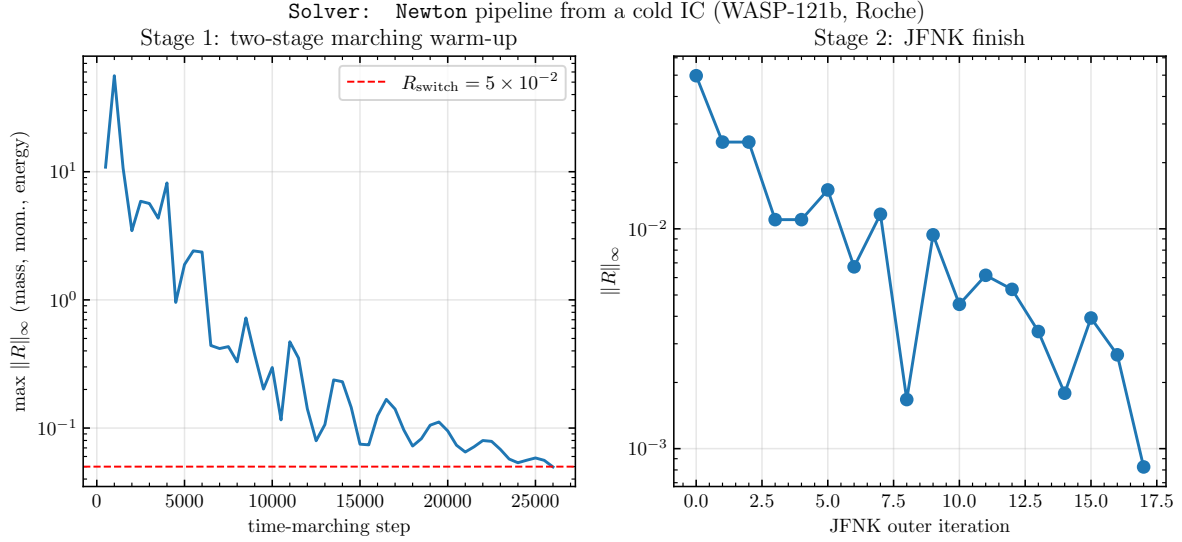


Figure 1: The Solver: Newton pipeline from a cold IC (WASP-121b, Roche). Left: maximum steady residual $\max \|R\|_\infty$ over the mass, momentum, and energy equations during the two-stage (Reconstruction scheme: PLM+WENO3) marching warm-up; the dashed line is the legacy residual hand-off threshold $R_{\text{switch}} = 5 \times 10^{-2}$ used when this figure was produced (the current default keys the hand-off on $du < 10^{-2}$ instead, at a comparable point on the warm-up). Right: residual vs. JFNK outer iteration during the Newton finish, down to Resid tol (10^{-3}).

2.8 Environment-variable hooks (diagnostics and expert runs)

A small set of *environment variables* expose diagnostic and expert entry points without touching `input.inp` (Table 1). They are read once at startup; when unset they have strictly zero effect, so normal runs are unaffected. All of them are diagnostics or validation aids — the only one used in a production workflow is the EXHALE_PTC family, which implements the “Newton from an existing state” recipe of §2.7 (and `examples/04_newton_from_state/`). Typical usage:

```
EXHALE_RESIDUAL=1 ./EXHALE.x # rank a "converged" state
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x # re-converge it
```

2.9 Metal cooling on/off comparison

Metal-line cooling is active *iff* a `metals.inp` is present (§3.); remove or rename it for a metals-off run. Because the metal abundances are read at run time, the executable is built once and the two cases differ only by whether `metals.inp` exists—no source edits or rebuild. To run the *same* planet both ways:

Variable	Effect
EXHALE_DUMP_IC=1	After Load IC? True, write the restored state to output/Hydro_ioniz.txt and output/Ion_species.txt and stop, <i>before</i> the first ionization-equilibrium solve re-equilibrates the species fractions (verifies a snapshot/restart round-trip).
EXHALE_RESIDUAL=1	Evaluate the steady residual R of the loaded state (WENO3, boundary conditions applied first), print the norms for each equation (mass, momentum, energy), and stop. The criterion-independent way to rank candidate “converged” states (a premature-dip state shows $\ R\ _E \sim O(10)$ despite $du < 10^{-3}$).
EXHALE_NEWTON_TEST=1	Self-test of the Newton infrastructure (pack_U/unpack_U round-trip and the frozen residual against the EXHALE_RESIDUAL values); prints and stops.
EXHALE_JAC_TEST=1	Verify the colored-FD banded Jacobian against a directional finite difference $Jr \simeq [F(Y+\epsilon r) - F(Y)]/\epsilon$; a correct band layout matches to $\sim 10^{-6}$, a wrong one is $O(1)$ off. Prints and stops.
EXHALE_PTC=1	Solve the <i>loaded</i> state directly with the steady solver (no marching), write output/, and stop. Requires Load IC? True; set Valve eps: $1.0\text{e-}4$ and Resid tol: $1.0\text{e-}3$ (§2.7). Default engine: banded PTC.
EXHALE_PTC_JFNK=1	With EXHALE_PTC=1: use the full JFNK solver (GMRES on the matrix-free J) instead of the banded PTC — the recommended engine.
EXHALE_PTC_DTAU0= $\langle v \rangle$	With EXHALE_PTC=1: override the initial pseudo-time step $\Delta\tau_0$ (default: the CFL Δt of the loaded state). A large value (e.g. 1.0) probes the pure-Newton regime directly.
EXHALE_PTC_NFIX= $\langle n \rangle$	With EXHALE_PTC=1: anchor the first n physical cells (frozen-base experiment; solves the wind on top of a fixed base).
EXHALE_DIAG_BASE=1	Write output/base_diag.txt (r, n, v, T , heating, cooling of the first few cells over the first steps) to trace IC-startup transients at the base.
EXHALE_FORCE_HYBRD1=1	Force every ionization-equilibrium solve to use MINPACK hybrd1, skipping the analytic-Jacobian Newton: lets the same binary produce both solvers’ outputs for an A/B validation (Update_EXHALE_solver.pdf §6).

Table 1: Environment-variable hooks. All are read once at startup and have zero effect when unset. The four diagnostic tests (top block) and EXHALE_PTC runs stop after printing/writing; they do not march.

```

make                                # build once
# metals ON (solar C/N/O from metals.inp):
printf 'CI 2.69e-4\nNI 6.76e-5\nOI 4.90e-4\n' > metals.inp
./EXHALE.x ; mv output output_metals_on
# metals OFF (no metals.inp):
rm -f metals.inp
./EXHALE.x ; mv output output_metals_off

```

The notebooks for each planet `metals_comparison*.ipynb` read the two folders and plot the temperature, velocity / mass-flux, H and metal ionization, and heating/cooling differences (cf. §7.).

2.10 Wind-AE warm-start initial conditions

A 1-D steady-state Parker-wind solver (a Fortran port of *Wind-AE*; Murray-Clay et al. 2009 / Broome et al. 2025) is included under `src/modules/wind_ae/` and can supply the initial condition for a hard-to-launch EXHALE run. There are two ways to use it; the underlying algorithm, ramp, and BC/IC are documented in `docs/wind_ae_solver.tex`.

In-process (recommended): IC mode: windae. Add the single line

```
IC mode: windae
```

to `input.inp` (with `Load IC? False`). On startup EXHALE maps the planet’s parameters to Wind-AE, ramps a shipped seed solution to the planet, solves the steady wind, and writes the IC onto the EXHALE grid (`output/*_IC.txt`), which is then loaded automatically — the whole input → Wind-AE solve → IC → run sequence is one `./run_EXHALE.sh` (one `EXHALE.x`) invocation. This is a *warm start*: EXHALE re-solves the ionization (and any metals) from the first step, so the Wind-AE IC need not be exactly self-consistent. Always finish with `Solver: Newton` for a quantitative $\dot{M}$ (§2.7). The shipped seed and spectrum live in `inputdata/windae_seed.csv` and `inputdata/windae_spectrum.inp`. The ramp runs in two stages, and the second one is what makes far-from-seed planets reachable. It first tries the *static-BC* ramp, which moves the system parameters (flux, gravity, star) while holding the seed’s base boundary conditions fixed — fast, and enough for a seed-adjacent hot Jupiter such as HD 209458 b. If that stalls, the seed is reloaded and the ramp is retried with the *self-consistent-BC* continuation, which re-converges the base $R_{\min}/T/\rho$, the sonic-point column density, and the bolometric/molecular layer turn-off as the parameters move (`wae_continuation.f90`). That is the default path, and with it the ramp converges the far-from-seed HD 189733 b as well as the seed-adjacent HD 209458 b (`examples/11_windae_ic` and `examples/12_windae_ic_hd209`). Only if *both* stages fail does the run stop with a message advising `IC mode: auto`. What the Wind-AE IC does not fix is EXHALE’s own HD 189733 b base-breathing instability, which limits the warm start there whatever the IC source.

Because a generated IC is an analytic profile that the marching loop has not touched, its du can start far below every threshold and would once have tripped a false convergence stop within a few steps. Both the du stop and the JFNK hand-off now require du to have been seen at or above their own threshold first, so only a descending crossing counts (§2.4). Measured on `examples/12_windae_ic_hd209` (2026-08-15): du at step 2 is 1.43×10^{-4} , neither trigger fires,

and the two arm at steps 23 and 222 on the first ascending crossing. The false stop is structurally gone for this IC mode; an $\dot{M}$ read off a du stop still carries its usual path-dependent spread, so the Newton finish remains the prescription.

Standalone generator: wind_ae_ic.x. The same solver builds an IC out of process:

```
make wind_ae_ic      # builds ./wind_ae_ic.x (separate from EXHALE.x)
./wind_ae_ic.x <input.inp> <seed.csv> <spectrum.inp> \
                <IC_dump_grid> <outdir> [rhoscale]
```

It reads an EXHALE `input.inp`, ramps from the seed, and writes `<outdir>/{Hydro_ioniz,Ion_species}_IC.txt` on the grid given by an `IC_dump.txt` from a prior EXHALE run with the same domain settings. Run EXHALE on the result with `Load IC? True + Solver: Newton`. (The standalone tool is independent of the in-process path above; either can be used.)

3. Input files

3.1 input.inp — the main parameter file

`input.inp` is read *line by line in a fixed order* for the core block (Table 2), followed by an *optional keyword block* (Table 3) whose lines are matched by keyword and may be omitted (older files without them keep the default behavior). A few lines are *conditional*: the spectrum-property line depends on `Spectrum` type, and the energy-band / X-ray-luminosity lines depend on `Use only EUV?`.

Table 2: `input.inp` core block (fixed order). Conditional lines are marked (*).

Label	Value	Meaning
Planet name:	string	Label for the run (cosmetic).
Log10 lower boundary number density [cm ⁻³]:	real (log ₁₀)	log ₁₀ n_0 , the H number density at the 1 μ bar base.
Planet radius [R _J]:	real (R_J)	Base radius R_p (the 1 μ bar level); the grid radius r is in units of this.
Planet mass [M _J]:	real (M_J)	Planet mass M_p (sets gravity / the Jeans parameter b_0).
Equilibrium temperature [K]:	real (K)	Base temperature T_0 .
Orbital distance [AU]:	real (AU)	Star–planet separation a (sets the XUV flux $\propto a^{-2}$ and the tidal/Roche potential).

Table 2, continued

Label	Value	Meaning
Escape radius [R_p]:	real (R_p)	Inner radius of the constant-momentum convergence window; must lie <i>inside</i> the domain ($< r_{\max}$). For L1-truncated Roche runs use a value below the lobe.
He/H number ratio:	real	Helium abundance $n_{\text{He}}/n_{\text{H}}$; > 0 enables He chemistry.
2D approximate method:	see text	Day–night averaging of the XUV flux and $\dot{M}$: Rate/2 + Mdot/2 (dayside, $\xi = \frac{1}{2}$), Mdot/4 (full-sphere), Mdot, or alpha (then a trailing number sets the rate-correction a_τ).
Parent star mass [M_sun]:	real ($M_\odot$)	Stellar mass $M_\star$ (tidal potential, Roche-lobe size).
Spectrum type:	Power-law Monochromatic Load	Shape of the incident XUV SED.
(*) Power-law index: / Photon energy [eV]: / Spectrum file:	real / real / path	Property of the chosen SED: power-law slope E^{PLind} , the monochromatic energy, or a numerical SED file. With metals, mind the required wavelength coverage (§3.5).
Use only EUV? True/False:	logical	True: EUV band only. False: include X-rays (adds the band’s third energy and an X-ray luminosity line).
(*) [E_low,E_mid,E_high] = [. . .]	reals (eV)	Band edges: two values (EUV only) or three (with X-rays).
(*) Log10 of X-ray luminosity [erg/s]: Log10 of EUV luminosity [erg/s]:	real (log ₁₀) real (log ₁₀)	Only if X-rays are included; log ₁₀ L_X . log ₁₀ L_{EUV} (with a , sets the incident XUV flux).
Grid type:	Uniform Stretched Mixed	Radial grid spacing. Mixed (fine near the base, stretched outward) is the usual choice.
Numerical flux:	HLLC ROE	Riemann solver for the hydrodynamic fluxes.

Table 2, continued

Label	Value	Meaning
Reconstruction scheme:	PLM WENO3 PLM+WENO3	Spatial reconstruction order and staging. PLM or WENO3 runs a single stage (only the first <code>du_th</code> value is used as the convergence threshold); PLM+WENO3 selects the two-stage mode (march in PLM to the first <code>du_th</code> value, then switch to WENO3 and converge at the second).
Include He23S?:	logical	Include the metastable He 2^3S triplet (needed for the He 10830 line). <i>On by default</i> , and the only line of the core block that may be omitted: a file without it gets the triplet. False is the deliberate opt-out, used to compare against the atomic helium limit. Forced off when the gas carries no helium.
Load IC?:	logical	Restart from the previous run's output/ <code>*_IC.txt</code> (warm start).
Do only PP:	logical	Skip the time loop and only run the post-processing on existing output.
Force start:	logical	Force the first 1000 iterations regardless of the convergence test (useful when the IC is close to a fixed point).

Table 3: `input.inp` optional keyword block (any line may be omitted; defaults restore the standard behavior).

Keyword	Value	Meaning
Domain mode:	Spherical	If present and Spherical: pure planetary potential $-b_0/r$, domain extended to Outer radius. Absent (default): full Roche (tidal) potential; outer boundary at the Hill/L1 radius unless Outer radius is given (§1.3).

Table 3, continued

Keyword	Value	Meaning
Outer radius [R_p]:	real > 1 (R_p)	Outer boundary (e.g. 10–15 R_p). In Spherical mode it sets the pure- $-b_0/r$ domain; in Roche mode (no Domain mode line) it extends the domain past L_1 while <i>keeping</i> the tidal potential (Yan/Huang-style; pair with IC mode: auto).
Base grid [dr,cells]:	real (R_p), int (def. 2.0e-4 50)	Resolution of the uniform region of the Mixed grid: cells uniform cells of size dr stacked on the base, then N -cells stretched cells out to $r_{\max}$. The product $\text{dr} \times \text{cells}$ is the extent of the uniform region ($0.01 R_p$ by default), so refining the base at fixed extent means dividing dr and multiplying cells by the same factor (5.0e-5 200 is the $4\times$ refinement). Requirement: dr must resolve the base density scale height $H = kT/(\mu g)$; below ~ 5 cells per H the scheme carries an undamped stationary $2dr$ entropy mode at the base (§2.5). Ignored by Grid type: Uniform and Stretched. At fixed Grid cells: a finer base leaves fewer cells for the stretched region; raise the two together to hold the upper stretch (§2.5).
Grid cells:	int ≥ 10 (def. 500)	Total number of computational cells N (ghost cells not counted). The default reproduces the value that used to be compiled in, so an input.inp without the key is unchanged. A restart (Load IC? True) must load an IC written at the same N ; the row count is checked and a mismatch is reported as such.

Table 3, continued

Keyword	Value	Meaning
Base ghost temperature:	isothermal continuous isothermal)	Temperature closure of the lower ghost cells (the density anchor ρ_{bc} is unaffected either way). isothermal pins $T_{ghost} = T_0$ through the fixed ghost pressure $n_{tot,bc} + \delta p_{bc}$ (legacy). continuous imposes $dT/dr = 0$: the ghost keeps the base composition but carries T_1 , the temperature of the first interior cell. Use it when the base cools well below T_{eq} , where the pin becomes a permanent contact discontinuity on the boundary (§2.6). Ignored if Hydrostatic base: True.
Max steps:	int (def. 10^6)	Hard cap on marching iterations (count_max); the run reports that it stopped without converging when the cap is reached. Lower it to bound an exploratory run, raise it for a high-gravity case whose base time step makes 10^6 steps too short (§2.4).
Coronal cutoff width:	real (def. 0.1)	Roll-off width w of the coronal-excitation guard below the 10^3 K CHIANTI fit floor: the coronal part of every metal-line coefficient is multiplied by $\exp(-x^2)$ with $x = (T_{floor}/T - 1)/w$, so smaller w cuts the extrapolated coronal cooling off more sharply. Since the ground-term fine-structure floors were solved in statistical equilibrium (§8.) the results no longer depend on w : the base-cell balance temperature of all four paper planets is identical to every digit printed over $w = 0.02$ – 1.2 , against a 500–650 K spread before. The default is kept at 0.1, and the guard is kept because extrapolating a fit below its fitted range is the wrong thing to do, not as a tuning knob (docs/coronal_cutoff_width.md, section 7).

Table 3, continued

Keyword	Value	Meaning
Base IR field:	True/False (def. False)	Let the infrared coolants of the molecular layer see the thermal radiation of the atmosphere below the base instead of emitting into vacuum. The lower atmosphere is taken to be black at those wavelengths and to radiate $B_\nu(T_0)$ over the sky fraction $1 - \sqrt{1 - (R_p/r)^2}$, and the cooling becomes the net rate, emission minus absorption of that field, so each band stops cooling at its own radiative-equilibrium temperature (576–642 K for the CI/OI fine-structure lines and 936 K for the H_3^+ bands, at $T_0 = 1140$ K and half-sky coverage). Applies to the eight ground-term fine-structure lines of CI, CII, NII, OI and to the H_3^+ infrared cooling; every other channel keeps the optically thin, no-incident-field limit. Off by default, and with no effect on an atomic run, where those channels carry no cooling (docs/lower_atmosphere_coupling.md).

Table 3, continued

Keyword	Value	Meaning
Stellar LW flux [erg/cm ² /s]:	real (def. 0)	Band-integrated stellar flux in the H ₂ Lyman–Werner bands (912–1110 Å, i.e. 11.2–13.6 eV) <i>at the planet’s orbit</i> . Switches on H ₂ photodissociation, $\text{H}_2 + h\nu \rightarrow \text{H} + \text{H}$, in the molecular network: an unattenuated rate $k_0 = 1.757 \times 10^{-7} F_{\text{LW}} \text{ s}^{-1}$ (Draine & Bertoldi 1996 calibration) times their eq. (37) self-shielding factor of the star-ward H ₂ column, plus 0.4 eV of heating per dissociation (Black & Dalgarno 1977). The band is a separate input because the code’s own energy grid does not carry it: a numerical SED is read only down to the 13.6 eV H I edge, and a power-law SED below 13.6 eV is an extrapolation of the XUV law rather than a stellar FUV spectrum. 0 (default) = no photodissociation, and the key is inert without Molecular chemistry: True. Writes output/Lyman_Werner.txt (docs/lower_atmosphere_coupling.md).
Stellar Teff [K]:	real (K)	Stellar T_{eff} for the diluted-blackbody Balmer continuum that photoionizes/heats H($n=2$). Enables the excited-H model when both this <i>and</i> Stellar radius are > 0 .
Stellar radius [R_sun]:	real ($M_{\odot}$ -radius)	Stellar radius: sets the Balmer-continuum dilution $R_{\star}/a$ here, and in EXHALE_transit.py it normalizes every transit depth and caps the impact-parameter integration at $R_{\star}/R_p$. Supply it for any run whose spectrum will be synthesized — absent, the transit script falls back to a built-in radius that need not match the system, and the depths scale accordingly.
Jlya RT file:	path	Read an external $\bar{J}_{\text{Ly}\alpha}(r)$ profile to pump H($n=2$) (sets jlya_mode=1).
Jlya escape-prob:	True	Compute $\bar{J}_{\text{Ly}\alpha}$ in-line each step by escape-probability RT (jlya_mode=2).

Table 3, continued

Keyword	Value	Meaning
Stellar Ly α flux [erg/cm ² /s]:	real	Incident stellar Ly α flux at the planet (the stellar source for jlya_mode=2).
Ly α stellar halfwidth [km/s]:	real (km/s)	Stellar Ly α line half-width; controls how deep the trapped stellar beam penetrates (default 70).
Ly α stellar boost [-]:	real	Bounded buildup cap for the trapped stellar Ly α beam (default 5).
Ly α absorbing bottom:	logical (def. False)	Treat the lower boundary of the jlya_mode=2 closure as a pure absorber (Huang et al. 2017: the H ₂ layer beneath the base destroys Ly α through accidental resonances): a downward wing-escape channel β_{bot} on the depth to the bottom joins the upward and Sobolev channels, lowering $\bar{J}_{\text{Ly}\alpha}$ (and n_2) in the bottom few scale heights only. No effect on jlya_mode 0/1.
Deexc heat:	logical (def. True)	Collisional de-excitation of H($n=2$) returns 10.2eV to the electrons. It is not a double count of the H I collisional-excitation cooling, which is the one-way coronal rate of Cen (1992) with no de-excitation term in it; most of the $n=2$ population is maintained by Ly α pumping, so the term is absorbed Ly α thermalized by a collision. Active only with the excited-H model; False reverts to the one-way ledger.
Transonic IC:	True	Use a transonic isothermal-wind initial condition (solved from the potential via the Bernoulli integral) instead of the hydrostatic IC; needed for deep-RLOF bases that do not launch from rest. Auto-enables Force start.
Newton solver:	logical (def. True)	Solve the ionization equilibrium with the analytic-Jacobian Newton method (with a MINPACK hybrd1 fallback); False reverts to the legacy hybrd1. Default is faster ($\sim 25\%$) and reproduces hybrd1 to $\sim 10^{-7}$.

Table 3, continued

Keyword	Value	Meaning
Brent solver:	logical (def. True)	Solve the post-process energy equation with a bracketing (Brent) root-finder that selects the physical root; False reverts to legacy hybrd1 + a 2 $\times$ -band reject.
du_th [PLM,WENO3]:	two reals	Convergence thresholds (e.g. 0.5 1.0e-3). Which are used is set by Reconstruction scheme: above. With PLM+WENO3 (two-stage) EXHALE marches in PLM until $du < \text{first value}$ (or a PLM stall), then switches to WENO3 and converges at $du < \text{second value}$; with single-stage PLM or WENO3 only the first value is used and the second is ignored. The PLM+WENO3 mode replaces the manual PLM $\rightarrow$ stop $\rightarrow$ Load IC+WENO3 restart of the official ATEs recipe.
Stall [tol,N]:	real, int	Stall-detector override (relative du change and window; defaults 10 ⁻⁶ , 2000).
CFL:	real (def. 0.6)	CFL-number override.
Hot Parker IC:	real (K)	Warm-seed initial condition: stable hydrostatic density with a warm T /ionization/velocity overlay.
IC mode:	word	Initial-condition family: cold (default), transonic, hot_parker (synonyms for the two keys above), auto — pick automatically from the cold sonic-point topology of the actual potential: an interior sonic point (deep-RLOF via the Roche L1 topology, or low-gravity boil-off, $r_c \simeq b_0/2c_0^2$ inside the domain) selects the transonic IC, otherwise the cold hydrostatic IC — or windae — build the IC in-process with the ported Wind-AE steady-wind solver (§2.10). No tunable threshold for auto; the choice and the escape parameter b_0 are logged. Explicit Transonic IC/Hot Parker IC keys override auto. See docs/auto_ic_design.md and, for windae, docs/wind_ae_solver.tex.

Table 3, continued

Keyword	Value	Meaning
Level tol:	real	Mass-flux <i>level</i> -stability gate: a converged/stalled stop additionally requires the mean $ \rho v r^2 $ over the wind window to change by less than this (relative) across the last <code>N_stall</code> steps. Guards against stopping on a transient <i>du</i> dip while the flux level still drifts (≤ 0 = off, default; §2.7).
Resid tol:	real	Residual-based convergence: stop when the steady residual $\max_k \max_j R_{j,k} / \max_j u_k $ falls below the value (monitored every 500 steps), <i>replacing</i> the <i>du</i> test (≤ 0 = off, default). <i>du</i> alone is blind to the energy imbalance (§2.7).
Time stepping:	Local	Pseudo-time steps in each cell (steady-state acceleration; not time-accurate). Reaches the deep state fast but leaves residuals in each cell; pair with a Newton finish.
Energy solver:	Explicit	Revert to the original forward-Euler source update (solver-isolation tests).
Valve eps:	real (code <i>v</i>)	Smooth base valve: ghost velocity $\frac{1}{2}(v + \sqrt{v^2 + \varepsilon^2})$ instead of $\max(v, 0)$. Differentiable at $v = 0$ (required by the Newton solver); the converged answer is insensitive to ε (10^{-4} vs. 10^{-5} tested identical). ≤ 0 = exact legacy valve (default).
Solver:	Newton [real]	Production steady-state solve: marching warm-up until the flux metric <i>du</i> (the radial spread of $\rho v r^2$) drops below the optional value <code>newton_du_switch</code> (default 10^{-2}), then the JFNK Newton finish to <code>Resid tol</code> (default 10^{-3}), then the standard final outputs and post-processing. Adopts <code>Valve eps: 10^{-4}</code> automatically if unset. See §2.7.

Table 3, continued

Keyword	Value	Meaning
Low-Mach damping:	real ε_4 [, real M_{th}] (def. -1, i.e. off; $M_{\text{th}} = 10^{-3}$)	Artificial dissipation of the $2\Delta r$ contact/entropy mode. The HLLC flux carries the middle wave at $S^* \simeq v$, so the dissipation it applies to the entropy family vanishes as the flow stagnates and that mode becomes marginally damped; where metal cooling stalls a shell (HD 209458 b with solar C/N/O, $r \simeq 1.02$, $M \sim 10^{-5}$) it grows to several times the local mean $ v $. This key adds a gated fourth-difference (Jameson, Schmidt & Turkel 1981) stress $\varepsilon_4 g(M) \rho_f \lambda_f (v_{j+2} - 3v_{j+1} + 3v_j - v_{j-1})$ to the numerical momentum flux, and its work $v_f D_p$ to the energy flux, with the gate $g = [\max(0, 1 - M^2/M_{\text{th}}^2)]^2$ exactly zero for $M \geq M_{\text{th}}$. It enters the numerical flux, so the marching loop and the JFNK steady residual see the <i>same</i> equation. Explicit stability requires $\varepsilon_4 < 1/(16\text{CFL})$ (0.10 at the default CFL); the classical range $1/64$ – $1/32$ works. $\varepsilon_4 \leq 0$ = off, bit-identical. See <code>src/modules/flux/low_mach_dissipation.f90</code> .
ATES_photoionization_rate:	True	Revert the He I (1^1S) photoionization cross section to the legacy ATES two-term fit (default: Verner+1996). Does <i>not</i> affect the He 2^3S Penning rate, which is always the temperature-dependent form (§8.).
Legacy_HHe_rates:	logical (def. False)	Revert the H/He case-B recombination and collisional ionization to the legacy ATES fits (Hui & Gnedin 1997 recombination; Abel+1997 / Hui & Gnedin 1997 collisional ionization). Default uses Badnell (2023) RR (+ He II DR) minus the Mao & Kaastra (2016) α_1 for case B, and Voronov (1997) collisional ionization (§8.). The free-free Gaunt factor and its ion net-charge weighting are always the van Hoof et al. (2014) form, independent of this switch.

Table 3, continued

Keyword	Value	Meaning
Secondary_ionization:	False True Immediate (def. True, staged)	Shull & van Steenberg (1985) secondary ionization by fast photoelectrons in the main loop: a photoelectron ejected with $E_0 > 40\text{eV}$ deposits only $f_{\text{heat}}(x)$ of its excess as heat and drives further HI / HeI ionizations, where x is the ionized fraction of the H+He nuclei. The default is <i>staged</i> (§2.4): the coupling switches on only after the wind first converges without it, then the run re-converges. False restores full photoelectron thermalization, bit-identical to the legacy path; Immediate applies it from step 0 (A/B tests only).
He_rec_coupling:	logical (def. True)	Couple He II $\rightarrow$ He I recombination radiation to HI ionization (Draine 2011 on-the-spot y/z ; §8.). Adds an HI photoionization rate $n_{\text{HeII}}n_e[z\alpha_B + y\alpha_1]$ with its photoelectron heating and shifts the He II recombination to $\alpha_B + y\alpha_1$; y is the local $\geq 24.6\text{eV}$ ground-capture share ionizing H, z the density-dependent cascade share. False = pure case B (photons lost locally, the legacy path; in the He 2^3S mode the singlet recombination is then neither case A nor case B).
He_H_charge_exchange:	logical (def. True)	The He $\leftrightarrow$ H charge-exchange pair (Koskinen 2013 rates via Huang 2023 Table 4): $\text{He}^0 + \text{H}^+ \rightarrow \text{He}^+ + \text{H}^0$ (endothermic) and the $\text{He}^+ + \text{H}^0 \rightarrow \text{He}^0 + \text{H}^+$ loss channel where neutral H dominates ($\sim 40\%$ of He II at $r \simeq 1.05$ on WASP-121 b, $< 1\%$ by $r \geq 1.2$). Applied by dedicated routines in <i>every</i> helium-bearing ionization system, independent of <code>cx_full</code> (which still gates the metal groups). False = legacy no-He-CX path, bit-identical.

Table 3, continued

Keyword	Value	Meaning
He_diffusion:	logical (def. False)	Enable diffusive separation of helium: transport the He element ratio $n_{\text{He}}/n_{\text{H}}$ with advection + molecular-diffusion settling, so He/H falls with altitude. Off = He/H frozen at the input He/H number ratio (§3.6).
He_Kzz:	real (cm^2s^{-1} , def. 10^9)	Eddy-diffusion coefficient used with He_diffusion: mixes He/H toward uniform below the homopause (Taylor+2025 K_{zz}).
He_ambipolar:	logical (def. True)	Ambipolar-corrected settling mass — in the ionized wind the polarization field lifts He ions, reducing settling ($\Delta m_{\text{eff}} = 3 - \frac{1}{2}(\bar{Z}_{\text{He}} - \bar{Z}_{\text{H}})$); False = neutral $\Delta m = 3$.
He_alphaT:	real (def. 0)	Thermal-diffusion factor α_T added to the settling drift ($\alpha_T \partial \ln T / \partial r$); 0 = off.
He_metal_diffusion:	logical (def. False)	Also diffuse each trace metal independently against H (own mass and binary D); heavier elements settle more. Off = metals frozen to H (§3.6).
Molecular chemistry:	True/False (def. False)	Add H_2 , H_2^+ , H_3^+ and HeH^+ to the coupled ionization equilibrium, with the H_2 photoionization opacity and heating and the H_3^+ infrared cooling. Requires He/H number ratio > 0 ; He_diffusion is refused. Turning it on also turns Molecular base on (see there). A metals.inp may be present — the metal stages are then solved in the same system (§3.7).
Molecular base:	True/False (def. False)	Remove the H_2 -bound particles from the base particle count ntot_{bc} using the chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$ partition, giving a lower base pressure and a heavier base μ with the species arrays still atomic (Tier 2a). Forced on by Molecular chemistry: True, because an atomic ntot_{bc} under a molecular base inflates the base temperature by the particle-count ratio and dissociates the layer the chemistry just built — so with the chemistry on the key is redundant (§3.7).

Table 3, continued

Keyword	Value	Meaning
Lower atmosphere:	none analytic vulcan [, real (R_f)] (def. none)	Lower-atmosphere pre-step run before the wind: generate the base composition over the 1-bar to 1 μ bar column — analytically (src/utls/run_lower.py) or by calling VULCAN (src/utls/vulcan_driver.py, hours on the first run, cached after) — and hand it over through base.inp (Tier 3). The second token is the 1-bar (transit) radius and is required. An existing base.inp is used as is and nothing is regenerated; the generator needs EXHALE_ROOT set to the install directory (§3.7).
Lower column:	real (R_f)	Tier-1 consistency <i>report</i> only: integrate the isothermal- T_{eq} hypsometric column with the chemical-equilibrium $H_2/H/He$ partition from the given 1-bar radius up to 1 μ bar and print the derived base radius, base fractions and μ next to the input Planet radius. Nothing is overridden (§3.7).
Hydrostatic base:	True (def. off)	Momentum-consistent base pressure. The legacy ghost pins p to $n_{tot,bc} + \delta p_{bc}$ at T_0 , which flattens dp/dr to ~ 0 at the base, whereas steady momentum balance needs $dp/dr = -\rho g$; on a strongly bound planet the resulting base-face momentum residual drives the breathing limit cycle. With this key the ghost pressure is a linear extrapolation of the interior gradient (cells 1 and 2), so dp/dr stays continuous through the base and can balance gravity; ρ stays anchored at ρ_{bc} and the base temperature floats slightly off T_0 . It fixes neither T nor p , so Base ghost temperature is ignored while it is on. Off = byte-identical legacy behavior.

Table 3, continued

Keyword	Value	Meaning
Base BC:	density pressure [, real (μbar)] (def. density)	What the lower boundary anchors. density (legacy) fixes the base number density at the n_0 of the core block. pressure (CETIMB-style) fixes the base pressure at the optional second token (default $1\mu\text{bar}$) and <i>derives</i> n_0 from $n_0 k_B T_0 n_{\text{tot},\text{bc}} = p_{\text{base}}$; the same isothermal ghost then pins p and T_0 and derives ρ . It is a re-parameterization of the same ghost, whose physical effect is a much less dense base.
Base velocity:	valve massflux (def. valve)	Velocity closure of the lower ghost. valve (legacy) one-way-valves the first interior velocity, $\max(v_1, 0)$ or its smooth form (Valve eps). massflux sets it from steady continuity instead, $v_{\text{ghost}} = F_c / (\rho_{\text{bc}} r^2)$ with F_c the wind's mass-flux constant, as CETIMB (Koskinen et al. 2013) does. It gives the cleanest base (the $\pm\text{m s}^{-1}$ base oscillation disappears) but converges $\sim 5\times$ slower, which is why the valve stays the default.
Shapiro filter:	real ε [, int n] (def. -1, i.e. off; $n = 4$)	Apply a weak Shapiro (1970) 1–2–1 low-pass filter $u_j += (\varepsilon/4)(u_{j-1} - 2u_j + u_{j+1})$ to the conservative variables every n steps, damping the gravity-unbalanced sound-wave (base-breathing) mode the same way CETIMB does. $\varepsilon \leq 0 = \text{off}$, and off is the default on purpose: with the filter on, an HD209458b cold-IC sweep left the transonic-wind saddle for an infall state ($v < 0$ everywhere) regardless of the velocity BC, while filter-off relaxed to a clean outflow. Enable it only for a case that actually breathes. Unlike Low-Mach damping it touches only the marching state, not the numerical flux, so the JFNK residual does not see it.

Table 3, continued

Keyword	Value	Meaning
Viscosity:	True/False, or real μ_0 [, real s] (def. False)	True adds the radial viscous force $(\nabla \cdot \tau)_r$ and its dissipation $q_\mu = \frac{4}{3}\mu(\partial_r v - v/r)^2$ with the calibrated $\mu(T) = \frac{4}{15}(m_H/k_B)\kappa(T)$ (monatomic Chapman–Enskog at Prandtl 2/3, tied to the Watson et al. 1981 conductivity) — the molecular transport that damps the near-base momentum imbalance the inviscid HLLC scheme lacks. A numeric value is instead a diagnostic power law $\mu = \mu_0 T^s$ in code units ($s = 0.7$ by default), kept for sensitivity scans. Off = byte-identical to the inviscid code.
Conduction:	logical (def. False)	Add heat conduction $r^{-2}\partial_r(r^2\kappa\partial_r T)$ with the Watson et al. (1981) atomic-hydrogen $\kappa(T) = 4.45 \times 10^4 (T/10^3 \text{ K})^{0.7} \text{ erg cm}^{-1} \text{ s}^{-1} \text{ K}^{-1}$. Off = byte-identical.
Resid norm:	vol Linf (def. vol)	Which norm the residual convergence test and the Newton trigger use. vol is volume-weighted, $\sum_j R_{j,k} V_j / \sum_j u_{j,k} V_j$ with $V_j = r_j^2 \Delta r_j$ — the fractional drift rate of the volume-integrated conserved quantity, and not inflated by the small near-base cells of the non-uniform grid, where $R \propto 1/\Delta r$. Linf restores the legacy max-over-cells norm.
Wind-AE seed:	path, or grid auto	Which converged Wind-AE solution IC mode: windae ramps from; the default is the shipped inputdata/windae_seed.csv. grid/auto instead picks the nearest solution in inputdata/windae_grid/ to this planet's $(M_p, R_p, F_{\text{XUV}})$, which is a shorter and more robust ramp than always starting from the single fiducial seed (§2.10).
Wind-AE seed out:	path (def. none)	Write the converged Wind-AE relaxation solution to this file as a reusable seed, e.g. to grow inputdata/windae_grid/. Omitted = nothing is written.

3.2 metals.inp — trace-metal abundances

`metals.inp` is *optional*: if absent, the run is metals-off (H/He only). If present, it lists the total elemental abundance n_X/n_H (summed over ionization stages; the equilibrium solver distributes each element over its stages) for any of C, N, O, Mg, Si, Ca, Na, K, S, Fe. Blank lines and # comments are ignored; any non-zero abundance turns metals on. Four control keys are also accepted: `pp_metals` (metal treatment in the advection post-process: 0=metal-free, 1=frozen at the converged equilibrium (default), 2=re-solve, not yet implemented), `cx_full` (1=full Huang+2023 charge-exchange network; default metal-H only), `cno_cool` (C/N/O line-cooling source: 1=closed-form fits to CHIANTI v11, including NI/NII cooling (default); 0=legacy AIOLOS analytic fits, which have no N cooling and deviate by 40–70% from CHIANTI for OI/OII in the wind-launch region — see `cooling_data/cno_cooling_comparison.ipynb`), and `eos_metals` (bulk-gas metal policy: 1=metals contribute their mass to the gas density and base density, their electrons to the free-electron density and the base pressure correction, and their nuclei to the total particle density — a $\sim 1\%$ mass effect at solar abundances, consistent with the charge balance inside the ionization solver (default); 0=legacy trace approximation, H/He-only gas budget). Solar reference values (Asplund+2009):

```
# metals.inp -- solar n_X/n_H (Asplund+2009)
C 2.69e-4 N 6.76e-5 O 4.90e-4
Mg 3.98e-5 Si 3.24e-5 Ca 2.19e-6
Na 1.74e-6 K 1.07e-7 S 1.32e-5 Fe 3.16e-5
pp_metals 1
```

3.3 opacity.inp — opacity model (optional)

`opacity.inp` selects the bound-free opacity model via `OPACITY_MODEL = A|C|P|T`: **A** analytic (default, the standard ATES cross sections); **C** constant (threshold σ times the factors for each species `OPA_CONST_HI/HEI/HEII/HEITR`); **P** pressure-broadened (Robinson & Catling $f(p) = 1 + a(p/p_0)^n$ via `OPA_PB_FACTOR=a`, `OPA_PB_EXPONENT=n`, `OPA_PB_PIVOT=p_0`); **T** tabulated (two-column .opa files per species, energy [eV] vs. σ [10^{-18} cm^2], given by `OPA_FILE_HI` etc.). Missing keys keep their defaults, so an absent `opacity.inp` is equivalent to model A.

3.4 Excited hydrogen and the Ly α pumping field

The non-LTE $H(n=2)$ model (the metastable $2s$ + resonance $2p$ levels, Christie+2013) feeds two things: in `EXHALE.x` the $H(n=2)$ photoionization adds a Balmer proton source and photoelectric heating to the energy/ionization balance; in `EXHALE_transit.py` the $n=2$ population sets the $H\alpha/H\beta$ transmission. The model is *off by default*; it switches on only when both `Stellar_Teff` and `Stellar_radius` are given (they supply the diluted-blackbody Balmer continuum that ionizes/heats $n=2$), so the standard run is unchanged.

The $2s/2p$ populations are driven by the **Ly α mean intensity** $\bar{J}_{\text{Ly}\alpha}(r)$, which pumps $1s \leftrightarrow 2p$. How $\bar{J}_{\text{Ly}\alpha}$ is obtained is selected by three mutually-exclusive options:

(mode 0) Parameterized — the default with no external input. If neither `Jlya_RT_file` nor `Jlya_escape-prob` is

present, $\bar{J}_{\text{Ly}\alpha}$ is the Huang et al. (2017, Eq. 6) estimate computed internally from the run’s own state,

$$\bar{J}_{\text{Ly}\alpha}(r) = \frac{0.1 F_{\text{LyC}}}{\Delta\nu_D(r)} \frac{1}{1 + \tau_{\text{Ly}\alpha}(r)},$$

where $\Delta\nu_D = \nu_0 \sqrt{2k_B T / \mu} / c$ is the local Doppler width, $\tau_{\text{Ly}\alpha}(r)$ is the *top-down* line-center Ly α optical depth integrated from the outside in through the H I column, and

$$F_{\text{LyC}} = \xi \frac{10^{L_{\text{EUV}}}}{4\pi a^2} \left(1 - e^{-\sigma_{\text{LyC}} N_{\text{HI}}}\right)$$

is the deposited Lyman-continuum flux (the incident EUV flux times the dayside factor ξ from 2D approximate method and the absorbed fraction set by the neutral-H column N_{HI}). This needs *no* external file — it is a self-contained, approximate “staging” field, adequate to get the coupling and signs right but not a true radiative-transfer solution.

(mode 1) External RT file — Jlya RT file: <path>. Read a tabulated $\bar{J}_{\text{Ly}\alpha}(r)$ profile (two columns, r/R_p and $\bar{J}$ in cgs) produced by a real Ly α radiative-transfer calculation (e.g. a Monte Carlo run); it is interpolated onto the grid and used directly.

(mode 2) In-line escape probability — Jlya escape-prob: True. Compute $\bar{J}_{\text{Ly}\alpha}$ every step from the current state by escape-probability radiative transfer (Neufeld/Harrington wing escape): a trapped *internal* field (recombination + collisional sources closed by the escape probability β) plus a trapped *stellar beam* set by Stellar Ly α flux (with the penetration controlled by Ly α stellar halfwidth and the buildup capped by Ly α stellar boost). The default closure has no lower boundary; Ly α absorbing bottom: True adds a downward loss channel into the H₂ layer beneath the base (a pure absorber, as in the Huang et al. 2017 Monte Carlo), which lowers $\bar{J}_{\text{Ly}\alpha}$ in the bottom few scale heights only. This is the most physical of the three but still an approximation; see Update_EXHALE.tex §12 for the full formulae.

EXHALE_transit.py uses the *same* logic for its H α /H β $n=2$ population: if a Jlya_file is set it is read (mode 1), otherwise the mode-0 parameterized estimate above is used (with F_{LyC} rebuilt from the run’s H I column and the stellar EUV). So a converged run with no Ly α input still yields H α /H β predictions, via the default field.

3.5 Spectral coverage for metal photoionization

Neutral metals have ionization thresholds *far below* the hydrogen Lyman limit (13.6 eV $\equiv$ 912 Å): the lowest is K I at 4.34 eV (2856 Å), followed by Na I, Ca I, Mg I, Fe I, Si I, S I, and C I (Table 4). To photoionize them at all, the modeled stellar spectrum must therefore reach these long (FUV/NUV) wavelengths — the EUV/X-ray band alone ($\lambda < 912$ Å, $E > 13.6$ eV) leaves every neutral metal un-ionized and artificially abundant, corrupting both the ionization balance and the predicted metal lines. How far the spectrum reaches depends on the SED type (Spectrum type in Table 2):

Power-law / monochromatic (Power-law). The code builds its own energy grid. When low-IP metals are active it *automatically* appends a sub-13.6 eV extension (20 log-spaced points) running from the H I edge down to the lowest ionization threshold among the active neutral metals, and evaluates the EUV power law there by *extrapolation* (the EUV normalization is anchored to L_{EUV} , which is defined over $[E_{\text{low}}, E_{\text{mid}}]$, then continued below E_{low}). No

extra input is needed — but the FUV/NUV flux that ionizes the metals is just the EUV slope extended downward, which real stars do *not* follow (their flux rises steeply into the NUV). Treat power-law metal ionization as indicative; load a real SED for quantitative metal work. When the He 2^3S triplet and low-IP metals are active together the sub-grid starts at the lowest threshold of either (4.80 eV for the triplet, 4.341 eV for K I), so one power-law run carries both.

Loaded SED (Load, Spectrum file:). The file's own energy/wavelength grid *is* the radiative grid (read verbatim, no extrapolation). The SED file must therefore **extend to at least the lowest threshold of your active metals** — e.g. to $\approx 2860 \text{ \AA}$ if K is included, or to whichever neutral metal in Table 4 has the lowest threshold in the active set. A SED truncated at the Lyman limit ($912 \text{ \AA}$) delivers *zero* ionizing flux to Na I / K I / Ca I / Mg I / Fe I / Si I / C I / S I, so those stages stay spuriously neutral.

Either way, the energy bands set by $[E_{\text{low}}, E_{\text{mid}}, E_{\text{high}}]$ and the luminosities L_{EUV}/L_X still describe the *ionizing* (H/He-driving) part of the spectrum above 13.6 eV; the metal extension is in addition to them.

Table 4: Neutral-metal photoionization thresholds (first ionization stage of each modeled element), as E_{th} and the corresponding wavelength $\lambda_{\text{th}} = hc/E_{\text{th}}$. The stellar SED must extend to at least the longest λ_{th} among the *active* metals to ionize them. Values are the VFKY96/NIST thresholds used in the code (species_table.f90).

Species	E_{th} [eV]	λ_{th} [$\text{\AA}$]	Note
K I	4.34	2856	lowest — sets the grid floor when K is present
Na I	5.14	2413	
Ca I	6.11	2028	
Mg I	7.65	1622	
Fe I	7.90	1569	
Si I	8.15	1521	
S I	10.36	1197	
C I	11.26	1101	
(H I)	13.60	912	Lyman limit (reference)
O I	13.62	910	$\simeq$ Lyman limit
N I	14.53	853	above the limit (already in the EUV band)

3.6 Diffusive separation of helium and metals

By default EXHALE is single-fluid: the composition is re-solved each step by *local* ionization equilibrium, which conserves the element ratios, so $n_{\text{He}}/n_{\text{H}}$ (and each metal/H) stays frozen at the input value at every radius. Setting He_diffusion: True adds a genuine transport equation for the helium element ratio $f = n_{\text{He}}/n_{\text{H}}$ — advection at the bulk velocity plus a molecular-diffusion *settling* drift of He relative to H,

$$\Phi_{\text{He}} = -D n_{\text{H}} \left(\frac{\partial f}{\partial r} + f G \right), \quad G = \frac{\Delta m g}{kT}, \quad D = \frac{1.52 \times 10^{18} (1/m_{\text{H}} + 1/m_{\text{He}})^{1/2} T^{1/2}}{n},$$

(Banks & Kockarts 1973), plus eddy diffusion K_{zz} (He_Kzz) on the gradient term. Helium then settles out relative to the escaping hydrogen and He/H *declines with altitude*, as in Taylor et al. (2025) and Xing et al. (2023). The equation is solved conservatively (implicit, tridiagonal) and the new split is written back into the composition conserving mass. Physics refinements: He_ambipolar (default on) reduces the settling in the ionized wind via the polarization field; He_alphaT adds thermal diffusion; He_metal_diffusion extends the same operator to each trace metal (heavier elements settle more, so they deplete faster than He).

The strength of the separation scales as (settling speed)/(wind speed): it is suppressed by fast escape and is stronger for heavier species. On a gentle escaper (HD 209458b) helium itself separates and the He 10830 line weakens; on a furious escaper (WASP-121b) helium is dragged out almost unfractionated while the heavy metals still settle strongly. Full physics, derivation, and a two-planet before/after comparison: docs/design_hehe_diffusion.md and docs/version_compare.{md, tex, pdf}. The flags are **default off**, so standard runs are byte-identical to before.

3.7 Lower-atmosphere connection (Tier 1–3)

Three opt-in layers connect EXHALE’s 1- μ bar base to the atmosphere below (survey, implementation, examples and figures: docs/lower_atmosphere_coupling.pdf). All are **default off**.

Tier 1 — analytic lower column (consistency report). Add

Lower column: 1.36 # radius of the 1-bar level [R_J], e.g. transit radius

and EXHALE integrates the isothermal- T_{eq} hypsometric column with the chemical-equilibrium $H_2/H/He$ partition (Koskinen et al. 2022) from 1 bar to 1 μ bar and *reports* the derived base radius (equilibrium value + a fully-atomic bracket), the base $H_2/H/He$ fractions and μ , next to the input “Planet radius”. Nothing is overridden. A large base q_{H_2} on a genuinely cool planet signals that the molecular physics below is required.

Tier 2a — molecular base (EOS only). Molecular base: True removes the H_2 -bound particles from the base particle budget (ntot_bc) using the same equilibrium fit — a lower base pressure and heavier base μ , with the species arrays still atomic. Crude on its own, and it is what Tier 2 turns on for itself, so the key is worth setting by hand only for a run that wants the base EOS without the chemistry.

Tier 2 — molecular chemistry.

Molecular chemistry: True # H_2 , H_2^+ , H_3^+ , HeH^+ in the coupled equilibrium

adds the four molecular species to the coupled ionization equilibrium (the Koskinen et al. 2022 reaction network), the H_2 photoionization opacity and heating (Yan et al. 1998 cross section), and the optically-thin H_3^+ infrared cooling (Miller et al. 2013 with their non-LTE factor). The output Ion_species*.txt gains four columns (H_2 H_2^+ H_3^+ HeH^+). It also turns Molecular base on automatically, so that key need not be given: with an atomic ntot_bc under a molecular base the base temperature is inflated by the particle-count ratio ($\sim 1.8\times$ for a fully molecular base) and thermally dissociates the layer the chemistry just built, so the parser couples the two rather than leaving the inconsistent combination reachable. Requires He; He_diffusion is refused by the parser. A metals.inp may be present: the trace

metals are then solved in the same system as the molecular network (System_HeH_mol_metals), because the two blocks share the free electron density and the metals are the dominant electron donors in the shielded molecular base. H_2 photodissociation in the Lyman–Werner bands is available and opt-in — add `Stellar LW flux [erg/cm2/s]`: with the band flux at the planet. Caveats: local *equilibrium* chemistry only (no molecular advection), $\gamma = 5/3$ retained. On hot Jupiters the result is a thin molecular base under a sharp $\text{H}_2 \rightarrow \text{H}$ front with an atomic wind above — the atomic assumption becomes a result rather than an input.

Converging a molecular run takes three keys an atomic run does not need: Solver: Newton $5.0\text{e-}2$, Resid tol: $1.0\text{e-}5$ and Max steps: 150000. The *du* descent of a molecular run is not monotonic, so the default 10^{-2} hand-off threshold can consume the whole step budget; and $\|R\|$ is set by the two or three cells just above the base, so a run stopped at the default Resid tol: $1.0\text{e-}3$ leaves the molecular layer still cooling (its energy residual there is still $0.7\times$ the local H_3^+ cooling rate). Marching alone never reaches that layer — its H_3^+ cooling time is $\sim 10^7$ CFL steps — so the Newton finish is not optional. The converged layer is much colder than the relaxation snapshot (~ 400 K on HD 209458 b, 190–300 K on a hot Uranus) and, without Base IR field, that collapse is the model's own steady state rather than a prediction; see docs/lower_atmosphere_coupling.pdf §7–§9 and item (G) of TO_BE_DONE.md.

Tier 3 — base.inp handoff (and VULCAN). An optional base.inp in the run directory overrides the base state before the derived constants are formed (each override is echoed; absent file = no-op):

```
T_base    1140.0    # [K]      -> overrides T0
r_base    1.402     # [R_J]    -> overrides "Planet radius" (1-ubar radius)
HeH_base  0.083     #          -> overrides the He/H number ratio
Kzz_base  1.0e9     # [cm2/s] -> sets He_Kzz (used by He_diffusion)
```

The pre-step can be invoked by *EXHALE itself*: with

```
Lower atmosphere: vulcan 1.36    # or: analytic 1.36 (arg = 1-bar radius R_J)
```

in input.inp, the code runs the VULCAN photochemistry (or the analytic column) at startup when no base.inp exists, then proceeds with the wind solve on the generated base — subroutine-style, with the first VULCAN run taking hours and later runs reusing the cached result in <run_dir>/vulcan_work/. Omitting the key disables the pre-step entirely. The VULCAN/ tree (obtained by setup_vulcan.sh; it ships FastChem inside it) carries its own citation requirements — see §3.8. Two generators are provided in src/Utils/:

- `run_lower.py <run_dir> --r1bar <R_J> [--guillot]` — the analytic column (isothermal T_{eq} , or the Guillot 2010 semi-grey $T(p)$); pure Python, no dependencies beyond numpy.
- `vulcan_to_base.py <out.vul> <run_dir> --mp <M_J> --r1bar <R_J>` — converts a **VULCAN** photochemical-kinetics output into the same base.inp, with the *photochemical* H_2/H dissociation state (on HD 189733 b photochemistry dissociates $\sim 11\times$ more H at 1 μbar than chemical equilibrium) and a hypsometric base radius using VULCAN's own $\mu(p)$ and $T(p)$.

Running VULCAN is handled automatically by the driver src/Utils/vulcan_driver.py, so no manual VULCAN setup is required. When the pre-step is requested it (i) obtains VULCAN if needed (via src/Utils/setup_vulcan.sh;

see §3.8) and copies the EXHALE/VULCAN/ tree into `<run_dir>/vulcan_work/` (once, then reused), (ii) builds a Guillot (2010) semi-grey $T(p)/K_{zz}$ atmosphere for the planet and selects a stellar UV spectrum from the shipped spectra by the host T_{eff} ($> 6800 \text{ K} \rightarrow 51 \text{ Eri}$; $5300\text{--}6800 \text{ K} \rightarrow \text{solar}$; $4300\text{--}5300 \text{ K} \rightarrow \epsilon \text{ Eri}$; cooler $\rightarrow \text{GJ 436}$), (iii) compiles FastChem if its binary is missing, (iv) runs VULCAN to steady state, and (v) converts the result to `base.inp`. The first run takes hours; the converged `<name>-photo.vul` is cached and reused on subsequent runs (regenerate with `--force`). The code root is taken from EXHALE_ROOT when set (for relocated installs). VULCAN's networks are H/C/N/O(/S): it supplies the H_2/H state and molecular mixing ratios, *not* the atomic-metal release (Na/Mg/Ca/Fe) — `metals.inp` stays user-supplied.

3.8 Obtaining VULCAN and FastChem

The lower-atmosphere pre-step uses the **VULCAN** photochemical-kinetics code, which ships the **FastChem** equilibrium-chemistry code inside it. These are third-party open-source codes and are *not* committed to the EXHALE repository (VULCAN/ is git-ignored). Fetch and prepare them with

```
src/utils/setup_vulcan.sh
```

which clones VULCAN into EXHALE/VULCAN/, applies the two modifications EXHALE needs, builds FastChem, and removes stale run products. EXHALE also runs this automatically the first time the vulcan pre-step is used and VULCAN/ is missing. Obtain updates from, and *cite*, the original sources:

- **VULCAN** — github.com/exoclimate/VULCAN (mirror github.com/shami-EEG/VULCAN). Cite Tsai et al. (2017, ApJS 228, 20) and Tsai et al. (2021, ApJ 923, 264).
- **FastChem** — github.com/NewStrangeWorlds/FastChem (shipped inside VULCAN). Cite Stock et al. (2018, MNRAS 479, 865), and Stock et al. (2022, MNRAS 517, 4070) for FastChem 2.

The modifications `setup_vulcan.sh` applies to the cloned VULCAN tree (i.e. the only changes needed if you set it up by hand):

File (in the VULCAN clone)	Modification
<code>make_chem_funs.py</code>	add <code>encoding=None</code> to the <code>np.genfromtxt(vulcan_cfg.com_file, ...)</code> call so the element-conservation check does not crash on Python 3
<code>vulcan_cfg.py</code>	set <code>use_photo = True</code> and <code>use_live_plot = False</code> (this file is the driver's template)
<code>output/*.vul</code>	delete (run products; regenerated in <code><run_dir>/vulcan_work/</code>)
<code>fastchem_vulcan/</code>	make to build the fastchem binary

On the EXHALE side (in this repository) the pre-step adds `src/utils/setup_vulcan.sh`, `vulcan_driver.py` and `vulcan_to_base.py`, and the `Lower atmosphere: key` in `input_read.f90/parameters.f90`.

4. Output files

All profiles are written to output/ as plain text, one row per radial cell, in *cgs* units. Two flavors exist: the equilibrium files (*.txt, written during the run) and the advection-corrected files (*_adv.txt, written once at the end by the post-processor, which re-solves the ionization balance including advection of the wind). The *_adv.txt files are the ones the analysis tools read.

Caveat — the _adv profile at a “breathing” base. The advection-corrected (*_adv.txt) *temperature and ionization* would otherwise develop a spurious spike / sawtooth in the dense lower atmosphere (typically $r \lesssim 1.1\text{--}1.2R_p$ for strongly Roche-filling planets such as WASP-121b). Two compounding causes, both confined to the base, are responsible:

1. *A breathing, inflowing base (a 1-D limitation).* When the planet nearly fills its Roche lobe the 1-D wind is subsonic / only marginally launched at L_1 , so the dense lower atmosphere does not settle into a clean monotonic outflow but into a slowly recirculating state with small *negative* (inflow) velocities and a stagnation point ($v = 0$) near the wind base. This is intrinsic to the spherically-symmetric (1-D) geometry — removing it would require a 2-D/3-D model — and is not a numerical bug.
2. *A spurious hot root in the metal-cooled post-processor.* The post-processor re-solves the energy/ionization balance in each cell *including* advection, assuming an outflow (each cell takes the next-inner cell as its upwind state). In the dense base the non-monotonic metal line-cooling curve gives the energy equation a second, spurious *hot* root that the Newton (MINPACK) solver can land on; through the upwind coupling one bad cell then cascades outward, producing the sawtooth and an overshoot spike (post_process_adv.f90). A guard rejects solves that fall outside a band around the converged equilibrium temperature, but mild spikes *within* that band can survive.

The post-processor guards against cause 2 directly: wherever the flow is *not* a clean outflow ($v \leq 0$, the breathing inflow base) it skips the advection correction and keeps the converged equilibrium ionization and temperature (post_process_adv.f90). This removes the spurious spike and sawtooth at the source — it breaks the upwind cascade — while preserving the genuine few-percent _adv correction in the outflow above the stagnation point. Cause 1 (the breathing base itself) is intrinsic to the 1-D geometry and remains, but it is now represented there by the smooth, physical equilibrium solution. The shipped _adv profiles are therefore clean, and the temperature panel of Figure 6 plots the _adv profile directly. (Velocity and density are *identical* in the eq and _adv files — the post-processor corrects only the ionization and temperature.)

Validity range of the advection correction. The inflow condition above is one of three under which the post-processor keeps the converged equilibrium ionization instead of correcting it. A cell is left at equilibrium when

1. $v \leq 0$ on either face (inflow): the upwind discretization has no upstream cell, and the residence time $\Delta r/v$ is negative;

2. $\text{Da} = (\Delta r/v)(\Gamma_{\text{HI}} + \alpha_{\text{HII}} n_e) > 100$: the gas relaxes to local ionization equilibrium many times over while it crosses the cell, so the equilibrium solution already solves the advection ODE and the correction can only add integration error;
3. $x_{\text{HII,eq}} < 10^{-6}$: the residuals carry the *neutral* fraction and report the ion density as $(1 - x_{\text{HI}})n_h$, so the ion fraction is resolved only to the solver's absolute tolerance $\sqrt{\epsilon} = 1.5 \times 10^{-8}$. In a shielded base with $x_{\text{HII,eq}} \sim 10^{-13}$ the returned value is numerical noise of arbitrary sign, which the upwind cascade would then carry outward as negative ion densities.

Conditions 1 and 2 are properties of the flow and 3 of the representation of the unknown, so none of them depends on whether metal cooling is active. In practice they trigger only in the dense base: in the four shipped planet runs no cell above $r = 1.04 R_p$ is affected, and the wind is corrected as before. The temperature solve applies condition 1 only. The count of cells left at equilibrium is printed at the end of the run. Full account: docs/postprocess_advection_validity.md.

4.1 Hydro_ioniz.txt / Hydro_ioniz_adv.txt

Thermodynamic profiles (7 columns). The dimensionless code variables are restored to cgs with the normalizations n_0 (base density), $v_0 = \sqrt{k_B T_0 / \mu}$, $p_0 = n_0 \mu v_0^2$, T_0 (base temperature), $q_0 = n_0 \mu v_0^3 / R_p$.

Col	Quantity	Unit	Note
1	r	R_p	radius (base = 1)
2	ρ (mass density)	$m_H \text{ cm}^{-3}$	mass density ($= \rho_{\text{code}} n_0$)
3	v	cm s^{-1}	radial velocity ($\times v_0$)
4	p	erg cm^{-3}	pressure ($\times p_0$)
5	T	K	temperature ($\times T_0$)
6	heating	$\text{erg cm}^{-3} \text{ s}^{-1}$	XUV photoionization heating ($\times q_0$)
7	cooling	$\text{erg cm}^{-3} \text{ s}^{-1}$	radiative cooling ($\times q_0$)

Column 2 is a *mass* density expressed in units of m_H per cm^3 (metals included under the eos_metals policy), not a number density; multiply by m_H to get g cm^{-3} .

4.2 Ion_species.txt / Ion_species_adv.txt

Ion number densities [cm^{-3}] (34 columns). Column 1 is r ; columns 2–7 are the H/He ions; columns 8–34 are the 27 metal-ion stages, in this order:

r ; HI HII HeI HeII HeIII HeI* ; CI CII CIII OI OII OIII NI NII NIII MgI MgII MgIII SiI SiII SiIII CaI
CaII CaIII NaI NaII KI KII SI SII FeI FeII FeIII

(HeI* is the metastable 2^3S triplet.) The ionization fraction of any element is the stage density divided by the sum over that element's stages. With Molecular chemistry: True four more columns (H2 H2p H3p HeHp, cm^{-3}) are appended after the metals; the # columns header always names the layout.

4.3 Cooling_breakdown.txt

Radiative cooling by channel [$\text{erg cm}^{-3} \text{ s}^{-1}$] vs. radius. Columns: 1= r , 2= T [K], 3= n_e [cm^{-3}], 4=total cooling; 5--10= the six H/He channels (recombination, collisional ionization, then collisional excitation split by absorber into H I — the Ly α -dominated H-line cooling —, He I (which also carries the He 2³S metastable terms) and He II, and finally bremsstrahlung); 11= the H₃⁺ infrared cooling (zero unless the run tracks the molecular network); 12--38= the 27 metal-line cooling channels (same ion order as Ion_species). The channels sum to column 4. Useful for identifying the dominant coolant vs. altitude (cf. Huang+2023 Fig. 10), and the H₃⁺ column is the coolant of a molecular base. The # col header line always names the layout.

4.4 Heating_breakdown.txt

Volumetric heating by channel [$\text{erg cm}^{-3} \text{ s}^{-1}$] vs. radius, the companion of Cooling_breakdown.txt and written unconditionally at the end of the run. Columns: 1= r [R_p], 2= T [K], 3= n_e [cm^{-3}], 4=total heating; 5--10= the photoionization heating split by *absorber* — H I, He I, He II, He 2³S, H₂, and the sum over the metal stages; 11= the photoelectric heating of the H($n=2$) population (the excited-H model); 12= the collisional de-excitation of H($n=2$), i.e. absorbed Ly α thermalized by a collision (Deexc heat); 13= the He-recombination coupling (He_rec_coupling); 14= He 2³S + H Penning ionization; 15= He 2³S + H₂ Penning ionization; and 16= H₂ Lyman–Werner photodissociation. The channels sum to column 4 on a molecular run as well as an atomic one: the molecular terms were added on 2026-08-15, and the closure was then measured at $\max |\sum_5^{16} / \text{heat}_{\text{total}} - 1| = 2 \times 10^{-15}$ on the Lyman–Werner regression case. The molecular channels (9, 15, 16) and the excited-H channels (11, 12) are identically zero when the run does not carry that physics. The run also prints the internal consistency of the photoionization split alone. The # col header line always names the layout.

4.5 Lyman_Werner.txt

H₂ photodissociation diagnostic, written when a molecular run carries a Stellar LW flux. 8 columns: 1 r [R_p], 2 T [K], 3 $x_{\text{H}_2} = 2n_{\text{H}_2}/n_{\text{H}}$, 4 n_{H_2} [cm^{-3}], 5 the star-ward H₂ column N_{H_2} [cm^{-2}], 6 the Draine & Bertoldi (1996) self-shielding factor f_{shield} , 7 the resulting dissociation rate k_{LW} [s^{-1}], and 8 the photodissociation heating [$\text{erg cm}^{-3} \text{ s}^{-1}$]. It shows directly how deep the band penetrates: $f_{\text{shield}} \rightarrow 1$ in the thin wind above the H₂ $\rightarrow$ H front, collapsing by orders of magnitude inside the self-shielded molecular base.

4.6 Excited_H.txt

H($n=2$) diagnostic (written when the excited-H model is on, i.e. stellar T_{eff} and radius are set). 16 columns: 1 r , 2 T , 3 n_{HI} , 4 n_e , 5 $\bar{J}_{\text{Ly}\alpha}$, 6 n_{2s} , 7 n_{2p} , 8 Balmer proton source S_p , 9 photoelectric heating, 10 de-excitation heating, 11 $\tau_{\text{Ly}\alpha}$, 12--14 ground-state HI photoionization / collisional-ionization / recombination rates (for the proton-budget comparison), 15--16 the internal and stellar contributions to $\bar{J}_{\text{Ly}\alpha}$.

4.7 EXHALE_setup.out and the mass-loss rate

EXHALE_setup.out is a human-readable setup report (planet/star parameters, the Jeans parameter, composition, numerics) plus the final result. The **steady-state mass-loss rate** is $\dot{M} = 4\pi\rho v r^2$ evaluated near the outer boundary, corrected by the 2-D-approximation factor from 2D approximate method ($\text{Rate}/2 + \text{Mdot}/2 \Rightarrow \times \frac{1}{2}$, $\text{Mdot}/4 \Rightarrow \times \frac{1}{4}$). It is reported as $\log_{10}\dot{M}$ [g s^{-1}]; divide by M_p and multiply by 1 Gyr to express it in $M_p \text{ Gyr}^{-1}$. The helper `exhale_io.mdot_log10 / mdot_Mp_per_Gyr` reproduce this.

5. Analysis tools

5.1 EXHALE_plots.py — live profile viewer

`python3 EXHALE_plots.py [--live N]` draws an 8-panel figure from `output/`: density, velocity, pressure, temperature, momentum $\log_{10}(\rho v r^2)$, the ion densities, and the H and He ionization fractions. With `--live N` it refreshes every N seconds (watch a run converge). If the `*_adv.txt` files exist they are overplotted (dashed) against the equilibrium profiles (solid). It also prints $\log_{10}\dot{M}$.

5.2 EXHALE_transit.py — transmission spectra

EXHALE_transit.py post-processes a converged run (`*_adv.txt + input.inp`) into transit transmission spectra and reports the line-center absorption depths (theoretical, instrument-convolved, and instrument+rotation-convolved) for **He 10830**, **Ly α** , **H α /H β** (from the non-LTE $n=2$ population, Christie+2013, pumped by the Ly α field of §3.4), and the metal resonance doublets **Mg II h&k**, **Ca II H&K**, and **Na I D** — the metals get the same full treatment as the H/He lines (both doublet components in one window, theoretical / instrument / +rotation spectra, written under the same `tpm_<line>` names), plus the validated single-component line-center and 4 Å-band depth table; a metals-off run skips them automatically. It supports two geometries via `geometry = 'spherical'` (Phase 5a) or `'triaxial'` (Phase 5b, the 3-D Roche-equipotential reconstruction in `roche_recon.py`).

Where the system parameters come from. Every planet and star parameter is taken from the `input.inp` of the run directory, so the spectra always describe the same system as the simulation. Each is resolved with the precedence

EXHALE_TRANSIT_* environment override > ./input.inp > built-in default,

so an explicit override still wins for an experiment. The labels are matched exactly as `input_read.f90` does, so line order is irrelevant.

Quantity	input.inp label	Override
R_p	Planet radius [R_J]	—
M_p	Planet mass [M_J]	—
T_{eq}	Equilibrium temperature [K]	—
a	Orbital distance [AU]	—
M_*	Parent star mass [M_sun]	—
L_{EUV}	Log10 of EUV luminosity	—
ξ (day–night)	2D approximate method	xi_override
R_*	Stellar radius [R_sun]	EXHALE_TRANSIT_RSTAR_RSUN
T_*	Stellar Teff [K]	EXHALE_TRANSIT_TSTAR
P_{spin}	from a, M_*, M_p (tidal locking)	EXHALE_TRANSIT_ROT

The planet spin period has no input.inp key: a close-in giant is tidally locked, so the default is the Keplerian orbital period $P_{\text{orb}} = 2\pi\sqrt{a^3/[G(M_*+M_p)]}$ built from the orbital distance and the masses already in input.inp. Only the instrument resolving powers, the wavelength windows, and the geometry switch remain script/environment settings — they describe the observation, not the system. At startup the script prints every resolved parameter together with its source (input.inp, env ..., or built-in default), and warns when Stellar radius/Stellar Teff are missing from input.inp; read that block first when a spectrum looks wrong. A missing Stellar radius is the one worth checking before anything else: transit depths are normalized by πR_*^2 and the chord integration stops at R_*/R_p , so the built-in fallback shifts every depth, and not by a single factor. Measured on the tutorial planet (a $1.13M_\odot$ star, so $R_* \simeq 1.155R_\odot$), the fallback returned a He 10830 depth of 22.8% where the correct radius gives 4.0%.

Outputs. Every line carries one key — He10830, Lya, Halpha, Hbeta, MgII, CaII, NaI — and both products of a line are named from that key and land in the run directory: the model curve `tpm_<line>.txt` (always written) and, when EXHALE_TRANSIT_FIG_PREFIX is set, the figure `<prefix><line>.png` together with its vector .pdf. A curve file has the columns `lambda[A]`, T_λ (theoretical), T_λ (instrument-convolved), T_λ (+rotation), so the excess absorption in percent is $(1 - T_\lambda) \times 100$; a metals-off run simply writes no metal curves. EXHALE_TRANSIT_SAVE_PREFIX decorates the curve name in the same way the figure prefix decorates the figure name; both are resolved against the run directory, so an absolute prefix writes elsewhere.

Step-by-step usage. EXHALE_transit.py reads output/Hydro_ioniz_adv.txt and output/Ion_species_adv.txt from path, which is the current directory unless EXHALE_TRANSIT_PATH names the run directory:

1. *Ensure the *_adv.txt files exist.* A normal converged run writes them automatically. To post-process an existing state (e.g. a Newton solution produced by the EXHALE_PTC hook, which writes only the eq files): copy `Hydro_ioniz.txt` → `Hydro_ioniz_IC.txt` and likewise for `Ion_species`, set `Load IC? True` and `Do only PP: True` in input.inp, and run `./EXHALE.x` once (seconds).
2. *Set what describes the observation,* if the defaults do not: the instrument resolutions and wavelength windows (EXHALE_TRANSIT_RES_*, _HE_LMIN/LMAX/N), the azimuthal sampling of the rotation integral, and geometry for a Roche run. The star and the spin period need nothing — they come from input.inp.

3. *Run* `MPLBACKEND=Agg EXHALE_TRANSIT_PATH=<run dir> python3 EXHALE_transit.py` (the Agg backend suppresses the interactive windows on headless machines). The script writes the `tpm_<line>.txt` curves into the run directory and prints the line-center absorption depths (theoretical / instrument-convolved / +rotation) and the metal-line summary; add `EXHALE_TRANSIT_FIG_PREFIX=tpm_` to save the figures as well.

Worked examples (the converged Newton solutions of WASP-121b and HD189733b, with all four H/He spectra and the metal lines): `backup/regression/tpm_wasp/` and `backup/regression/tpm_hd189/`, collected in the notebook `backup/regression/steady_solver_results.ipynb`. Figure 2 shows the four H/He spectra `EXHALE_transit.py` saves for the WASP-121b Newton solution of `backup/regression/tpm_wasp/` (run with `Include He23S? True`, so the He I 10830 line is populated).

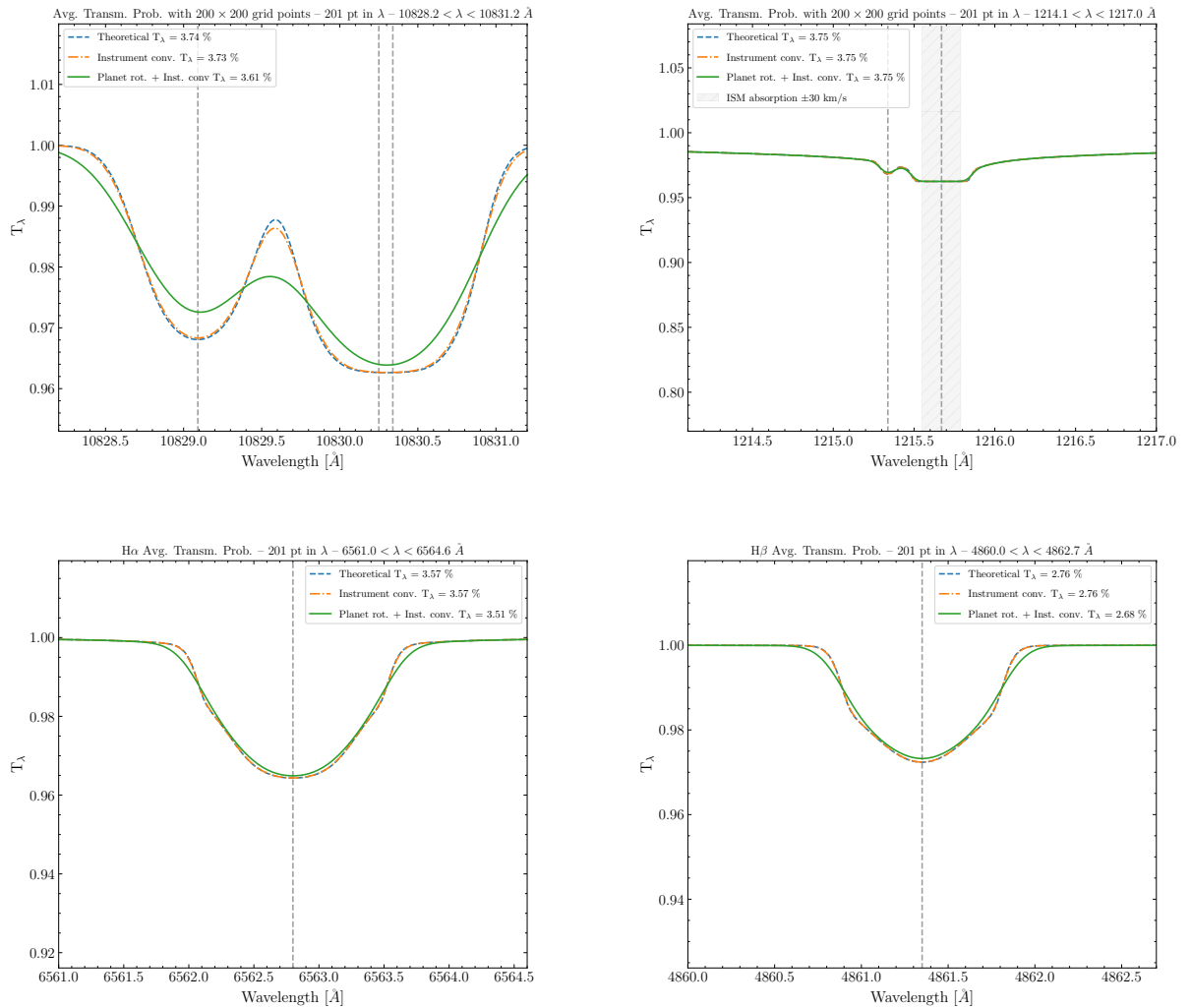


Figure 2: `EXHALE_transit.py` output spectra for WASP-121b (Roche, Newton solution with He 2³S on; `backup/regression/tpm_wasp/`): He I 10830 (top left), Ly α (top right), H α (bottom left), and H β (bottom right). Each panel shows the theoretical line, the instrument-convolved line, and the instrument+rotation-convolved line as produced by the script.

Transit-depth conventions (important). Huang et al. (2023) quote transit depths as the *effective transit radius* R_{eff}/R_* , not a percent absorption; convert from the code's line absorption fraction h via $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$. The metal NUV lines (Mg II) are quoted in a 4 Å bin, while the optical lines (Ca II, H α , Na) are line-center; line-center vs. 4 Å can differ by $\sim 3\times$ for an optically-thick line, so always compare in the same definition.

5.3 exhale_io.py — named, physical-unit loaders

examples/exhale_io.py loads every EXHALE output file into named arrays in physical units, so analysis code need not remember column orders (Table 5). The central object is the Run container: `load_run(outdir, inputfile)` fills it with the hydro profiles (r [R_p], n [cm^{-3}], v [cm s^{-1}] with a `v_kms` shortcut, p , T , heat, cool), the full ion-density dictionary `run.ion` (the 6 H/He plus 27 metal stages of §4.), and the parsed `input.inp` in `run.inp`. By default the `_adv` (advection-corrected) files are read; pass `adv=False` for the equilibrium snapshot. Typical use:

```
import exhale_io as aio
run = aio.load_run('output', 'input.inp')      # _adv profiles by default
print(aio.mdot_log10(run))                     # log10 Mdot [g/s], 2D factor incl.
print(aio.mdot_Mp_per_Gyr(run))               # same in M_p/Gyr
xh = run.x_ion(['HI', 'HII'])                 # stage fractions of one element
plt.plot(run.r, run.T)
plt.plot(run.r, xh['HII'])
cool = aio.load_cooling('output/Cooling_breakdown.txt')
plt.semilogy(cool['r'], cool['chan']['FeII']) # one cooling channel
```

The notebook examples/EXHALE_analysis.ipynb uses these loaders to plot the profiles and the parameter-difference comparisons of §6., and examples/make_figures.py (which generates every figure in this manual) is a further worked example.

5.4 exhale_to_lart.py — LaRT Ly α input from an EXHALE run

EXHALE_transit.py estimates the Ly α mean intensity $\bar{J}_{\text{Ly}\alpha}$ with the simple Huang et al. (2017) prescription (§3.4). For a proper treatment of the resonant Ly α radiative transfer that sets the $n=2$ hydrogen population (and hence H α), the converged atmosphere can be post-processed with the Monte Carlo code **LaRT** (Seon & Kim 2020; Seon et al. 2022). examples/exhale_to_lart.py builds a complete LaRT spherical-illumination input set directly from an EXHALE run.

Given a run directory it loads the `_adv` profiles (via `exhale_io.py`) and `input.inp`, and writes

- `dens_profile.txt` — $r[R_p]$, $n_{\text{HI}} [\text{cm}^{-3}]$ (the Ly α scatterer is ground-state H);
- `temp_profile.txt` — $r[R_p]$, $T [\text{K}]$;
- `velo_profile.txt` — $r[R_p]$, $v_r [\text{kms}^{-1}]$;
- `line_profile.txt` — the incident stellar Ly α profile, a double-Gaussian with peaks at $\pm m$ and Gaussian width s (defaults $m = 74$, $s = 49 \text{ kms}^{-1}$, the Huang et al. 2017 / Yan et al. 2022 values; pass `--fwhm` to read s as FWHM);

Function	Returns
<code>load_run(outdir, inputfile, adv=True)</code>	a Run: hydro profiles + <code>run.ion</code> dict + parsed input.
<code>Run.x_ion([stages])</code>	ionization fractions of one element, e.g. $x_{\text{ion}}(['\text{HI}', '\text{HII}'])['\text{HII}'] = n_{\text{HII}}/(n_{\text{HI}} + n_{\text{HII}})$.
<code>load_hydro(path)</code>	one <code>Hydro_ioniz(_adv).txt</code> as a dict (<code>r, n, v, p, T, heat, cool</code>).
<code>load_ions(path)</code>	(<code>r, {ion: density}</code>) from <code>Ion_species(_adv).txt</code> (all 33 stages).
<code>load_cooling(path)</code>	<code>r, T, ne, cool_total</code> + chan dict of cooling in each channel (6 H/He channels <code>rec/coll_ion/coex_HI/coex_HeI/coex_HeII/brems</code> , the H3p infrared channel, + 27 metal-line channels).
<code>load_excited_H(path)</code>	<code>Excited_H.txt</code> as a dict (<code>n2s, n2p, Jlya, tau_lya, Sproton, Hpe, . . .</code>).
<code>load_lyman_werner(path)</code>	<code>Lyman_Werner.txt</code> as a dict (<code>x_H2, NH2, f_shield, k_LW, heat_LW</code>).
<code>read_input(path)</code>	<code>input.inp</code> as a dict (<code>Rp_RJ, Mp_MJ, a_AU, HeH, LX, LEUV, . . .</code> + raw lines).
<code>mdot_log10(run, j_from_top=20)</code>	$\log_{10} \dot{M}$ [g/s]: $4\pi\rho v r^2$ near the outer boundary <i>including</i> the 2D approximate method factor (mirrors <code>EXHALE_main.f90</code> ; see §4.).
<code>mdot_Mp_per_Gyr(run)</code>	the same in units of $M_p \text{ Gyr}^{-1}$.
<code>constants</code>	<code>RJ, MJ, Msun, AU, GYR</code> (cgs).

Table 5: `examples/exhale_io.py` public interface. Run analysis scripts either from `examples/` or after adding it to the Python path (`sys.path.insert(0, '<repo>/examples')`).

- `<name>_lya.in` — the LaRT namelist (`geometry='spherical_atmosphere', source_geometry='stellar_illumination'`);
- `run.sh` — a multi-host `mpirun` launcher.

The length unit is set to R_p (`distance2cm =  $R_p$` in cm), so all LaRT radii are in R_p and match the EXHALE grid; the star–planet geometry (`distance_star_to_planet =  $a/R_p$` , `stellar_radius =  $R_*/R_p$`) is read from `input.inp`. Because the stellar illumination breaks spherical symmetry, the scattering rate is requested as a cylindrical 2-D grid (`geometry_JPa = 2`): $P_\alpha(\rho, z)$ is axisymmetric about the star–planet axis, which is the form needed to build the 2-D $n_{2p}(\rho, z)$ population for the H α transit. To populate P_α , build LaRT with `make CALCPnew=1` (producing `LaRT_calcP.x`).

```
# build the LaRT input from a converged EXHALE run, then run it
python examples/exhale_to_lart.py WASP-52b/fxuv0p25_he98 \
    -o WASP-52b/LaRT_lya --name WASP-52b \
    --m 74 --s 49 --rmax 10 --ngrid 201 --nphotons 1e7 \
    --hosts lart4,lart3,lart2
cd WASP-52b/LaRT_lya && ./run.sh
```

It is also importable: `import exhale_to_lart as e2l; e2l.build(run_dir, outdir, m_kms=74, s_kms=49, geometry_jpa=2, ...)`.

6. Worked examples

6.1 A first run: a generic hot Jupiter

The directory `examples/tutorial/` contains a minimal, self-contained model of a generic HD 209458b-like hot Jupiter: a spherical domain out to $10R_p$, solar metals, EUV+X-ray power-law SED. The full `input.inp` is:

```
Planet name: demoHJ
Log10 lower boundary number density [cm^-3]: 12.70
Planet radius [R_J]: 1.38
Planet mass [M_J]: 0.69
Equilibrium temperature [K]: 1320.0
Orbital distance [AU]: 0.047
Escape radius [R_p]: 2.00
He/H number ratio: 0.083
2D approximate method: Rate/2 + Mdot/2
Parent star mass [M_sun]: 1.13
Spectrum type: Power-law
Power-law index: -1.0
Use only EUV? False
[E_low,E_mid,E_high] = [ 13.60 , 123.98 , 1.24e3 ]
Log10 of X-ray luminosity [erg/s]: 27.0
Log10 of EUV luminosity [erg/s]: 29.30
Grid type: Mixed
```

```

Numerical flux: HLLC
Reconstruction scheme: PLM+WENO3
Include He23S? True
Load IC? False
Do only PP: False
Force start: False
Domain mode: Spherical
Outer radius [R_p]: 10.0
du_th [PLM,WENO3]: 0.5 1.0e-3
Solver: Newton
Stellar radius [R_sun]: 1.155

```

The `Reconstruction scheme: PLM+WENO3` line enables the two-stage reconstruction (with the two `du_th` values as the PLM and WENO3 thresholds), and the `Solver: Newton` line adds the steady-state Newton finish (§2.7), so the run converges to the true steady state without manual restarts. With `metals.inp` present (solar abundances, §3.), run `./EXHALE.x` in that directory. Figure 3 shows the converged profiles read back with `exhale_io`: the wind heats to $\sim 10^4$ K, the velocity rises monotonically outward (transonic outflow), and hydrogen ionizes across a front near $1.2\text{--}1.4R_p$.

6.2 Turning physics on and off

The same model is re-run with single-line changes to illustrate each option: *metals off* by removing `metals.inp`; *Roche/RLOF* by removing the `Domain mode/Outer radius` lines (the default truncates at L1); the *atomic-helium limit* by setting `Include He23S? False` (the triplet is on by default, so this is the change that removes it); the *H α /Balmer* physics by adding `Stellar Teff/Stellar radius`; and the *transonic IC* by adding `Transonic IC: True` for a near-unbound (deep-RLOF) base; and *diffusive He/H separation* by adding `He_diffusion: True` (extend it to the metals with `He_metal_diffusion: True`; §3.6). The comparison studies below quantify the effect of the first two.

6.3 Ready-made example configurations (examples/)

For every solver/physics combination there is a ready-to-run folder under `examples/` (paths relative to the EXHALE root). Folders 01–11 are the *same* planet (HD189733b) so that the effect of each option can be isolated; 12 and 14–16 use HD209458b (where the Wind-AE warm start, diffusion and molecular chemistry, with and without metals, are validated), and 13 spans four planets. Each folder is self-contained: `cd` into it and run the repo-root binary (`././EXHALE.x`); outputs land in the folder’s own `output/`. The “Added / changed lines” column of Table 6 is relative to each example’s natural base: the legacy `01_legacy_marching/input.inp` for the solver examples 02–04 (and Wind-AE 11/12), and the recommended-solver `03_newton/input.inp` (two-stage PLM+WENO3 + `Solver: Newton`) for the physics examples 05–10, which all inherit that solver and add only their signature option on top. A diff against the appropriate base shows exactly what an option adds; all configurations have been smoke-tested. The metastable helium triplet is on by default in the code, but the ladder holds `Include He23S? False` on purpose — in both bases

Tutorial run: generic hot Jupiter (spherical, H/He + metals)

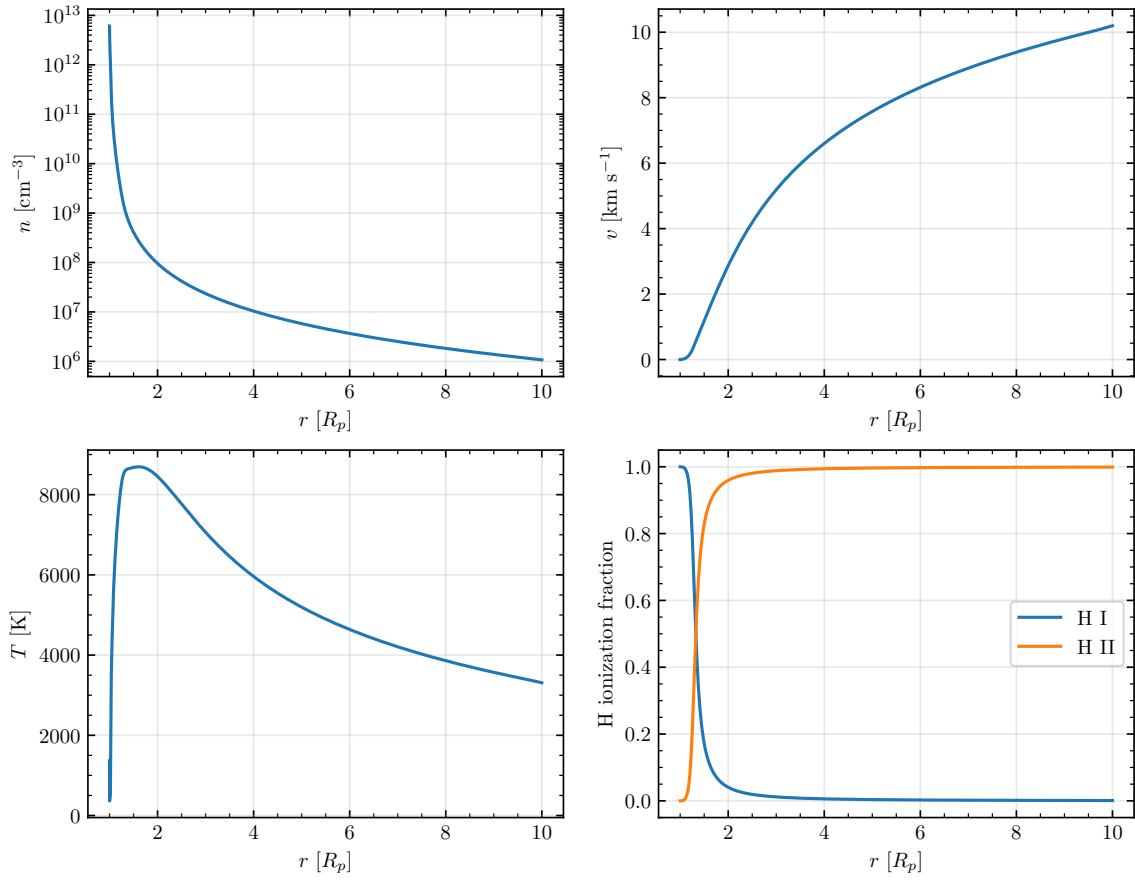


Figure 3: First-run output of the generic hot-Jupiter tutorial: density, velocity, temperature, and the H ionization fraction vs. radius.

and in every rung whose signature option is something else (02, 04, 05, 09–12) — so that atomic helium is the baseline and 06_he23s remains a one-line difference. Outside the ladder (tutorial/, 13–16, the planet folders) the triplet is on. 11_windae_ic (HD189733b) exercises the self-consistent-BC continuation: its Wind-AE ramp converges and writes the IC, though EXHALE’s own time-integration of HD189733b is limited by a separate base-breathing instability (§2.10). 12_windae_ic_hd209 is the seed-adjacent HD209458b case. 14_diffusion and 15_molecular turn on the He/metal diffusive separation (§3.6) and the full molecular network (§3.7) respectively, on HD209458b, and 16_molecular_metals adds a metals.inp to the latter.

Table 6: Ready-made example configurations under examples/ (01–11 are HD189733b; 12, 14–16 are HD209458b; 13 spans four planets). The Wind-AE IC ramp converges for both of the planets that use it — HD 189733 b (11) and HD 209458 b (12) — and 11, being far from the seed, additionally exercises the self-consistent-BC continuation and the molecular-layer turn-off. Metals on/off is a runtime switch: any folder becomes metals-on by copying a metals.inp into it, molecular folders included — that is what 16_molecular_metals is.

examples/...	Demonstrates	Added / changed lines
01_legacy_marching	classic ATES v2 baseline: PLM marching, H/He only, Roche domain, <i>du</i> -based stop	(none — baseline)
02_two_stage	two-stage PLM→WENO3 marching (§2.4)	Reconstruction scheme: PLM+WENO3, <i>du_th</i> [PLM,WENO3]: 0.5 1.0e-3
03_newton	recommended default: two-stage warm-up + Newton finish (§2.7)	+ Solver: Newton
04_newton_from_state	Newton re-convergence of an existing state (no marching; needs output/*_IC.txt, see the folder README)	Load IC? True, Valve eps: 1.0e-4, Resid tol: 1.0e-3; run via the EXHALE_PTC hook
05_metals	trace-metal cooling, solar C/N/O/Mg/Ca/Na/Fe (§3.)	metals.inp present
06_he23s	He 2 ³ S metastable level (HeI 10830); the one rung that leaves the ladder’s atomic-helium baseline	Include He23S? True
07_balmer_lya	non-LTE H(<i>n</i> =2) + Ly α pumping (H α /H β , §3.4)	+ Stellar Teff/radius, Deexc heat, Jlya escape-prob, Stellar Lya flux/halfwidth/boost
08_full	everything on (= the HD189733b/planet folder + Newton); feeds EXHALE_transit.py	07 + metals.inp

Table 6, continued

examples/...	Demonstrates	Added / changed lines
09_spherical	spherical domain instead of the default Roche/L1 truncation (§1.3)	Domain mode: Spherical, Outer radius [R_p]: 10.0
10_warm_seed_ic	warm-seed (hot Parker overlay) initial condition	Hot Parker IC: 10000
11_windae_ic	in-process Wind-AE IC for HD189733b (§2.10): far from the seed, so the self-consistent-BC continuation (base-BC re-convergence + molecular-layer turn-off) is exercised; the Wind-AE ramp converges and writes the IC. EXHALE's own HD189733b base-breathing instability (separate from the IC) then limits the warm start	+ IC mode: windae, Solver: Newton; symlinks inputdata/ for the shipped seed
12_windae_ic_hd209	in-process Wind-AE warm-start IC that <i>works</i> : HD209458b (not HD189733b) is close to the shipped seed, so the ramp converges and EXHALE warm-starts cleanly (spherical $10R_p$)	HD209458b params + Domain mode: Spherical, IC mode: windae, Solver: Newton; symlinks inputdata/
13_lower_atmosphere	lower-atmosphere connection for four planets (HD209458b, HD189733b, WASP-121b, WASP-52b; §3.7): an analytic $1\mu\text{bar}$ base column plus a base.inp hand-off (isothermal or Guillot $T(p)$ generator); see the folder README	Lower column: <R_1bar>; driver-generated base_iso.inp / base_guillot.inp
14_diffusion	diffusive separation of He and metals (HD209458b; §3.6): the He/H ratio declines with altitude and each trace metal settles independently, re-shaping He I 10830	HD209458b params + Include He23S? True, He_diffusion: True, He_metal_diffusion: True, He_Kzz: 1.0e9, He_alphaT: 0.0; metals.inp present

Table 6, continued

examples/...	Demonstrates	Added / changed lines
15_molecular	full molecular chemistry (HD209458b; §3.7): $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ in the coupled ionization equilibrium; a sharp $\text{H}_2 \rightarrow \text{H}$ front forms above a thin molecular base, the wind above it essentially atomic. Newton-converged from a cold start, re-run 2026-08-15 with the current code ($\ R\ = 6.105 \times 10^{-6}$ in 89 outer iterations): the front sits at $r = 1.0097 R_p$ and the molecular layer below it collapses to ~ 400 K	HD209458b params + Molecular chemistry: True, Solver: Newton 5.0e-2, Resid tol: 1.0e-5, Max steps: 150000 (no metals.inp here; He_diffusion is still refused)
16_molecular_metals	molecular chemistry <i>and</i> trace metals solved in one system (HD209458b): the $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ network and the metal ionization stages share the free electron density, which the metals dominate in the shielded molecular base. Converges only from a warm restart, and the seed matters: seed it with a 15 converged by <i>the same binary</i> (a state left by an older binary is a different seed and fails, info = 2), then expect info = 0 at Resid tol: 1.0e-5 and $\log_{10} \dot{M} = 9.65-9.67$. The cold path is closed: with metals on, the JFNK line search bottoms near $\ R\ \sim 10^{-3}$ (worst cell $r = 1.021$, momentum) and falls back to marching at every Low-Mach damping coefficient tried — off, 5×10^{-3} , 10^{-2} , 2×10^{-2} (measured 2026-08-15)	as shipped, 15_molecular + metals.inp (solar C/N/O) without 15's three convergence keys. For the warm restart put the seed run's Hydro_ioniz.txt/Ion_species.txt in output/*_IC.txt and use 15's three keys + Load IC? True + Low-Mach damping: 5.0e-3 (run 1.0e-2 as well and compare — see the folder README)

7. Comparison studies

7.1 Effect of metal-line cooling (metals on vs. off)

Trace metals add strong line cooling (and some photoionization heating and free electrons). Figure 4 compares the tutorial planet with and without metals. `inp`: metal-line cooling lowers the peak temperature by ~ 1800 K (acting as a thermostat that prevents the H/He-only over-heating), while the H ionization structure is barely changed. The effect is far stronger for hotter, more irradiated planets (e.g. WASP-121b, where metal cooling is essential to match the observed temperatures); the detailed cooling by channel budget is in Figure 5.

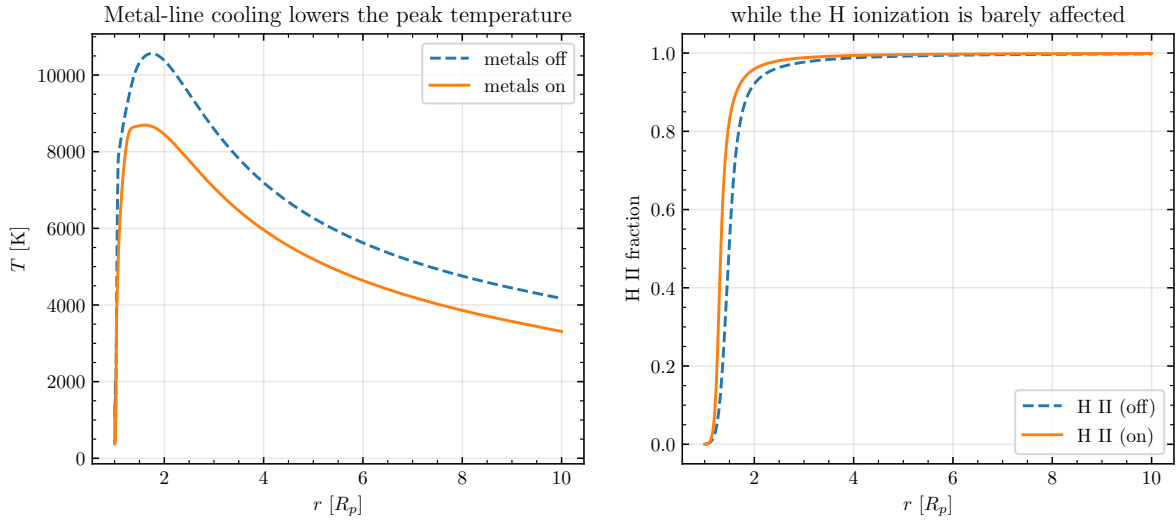


Figure 4: Metals on vs. off for the tutorial planet: metal-line cooling lowers the peak temperature (left) while the H ionization fraction is nearly unchanged (right).

7.2 Spherical vs. Roche (RLOF) and the mass-loss rate

Removing the `Domain` mode: `Spherical` line switches on the full Roche tidal potential, truncated at the L1 radius. Figure 6 compares a spherical model (Case A) and a Roche model (Case B) of WASP-121b: the tidal potential lowers the effective gravity toward the star, so the Roche wind accelerates far faster and the mass-loss rate rises by an order of magnitude ($\dot{M}$ from ~ 0.05 to $\sim 0.34 M_p \text{ Gyr}^{-1}$ here) — the “Roche-lobe overflow” enhancement.

7.3 Transmission spectra

`EXHALE_transit.py` turns a converged model into transit absorption. The metal resonance lines are the key NUV/optical diagnostics; Figure 7 compares the modeled effective transit radii R_{eff}/R_* (spherical Case A) against Huang et al. (2023). Recall the conventions of §5.2: Mg II is a $4 \text{ \AA}$ bin, Ca II/Na are line-center, and $R_{\text{eff}}/R_* = \sqrt{(R_p/R_*)^2 + h}$.

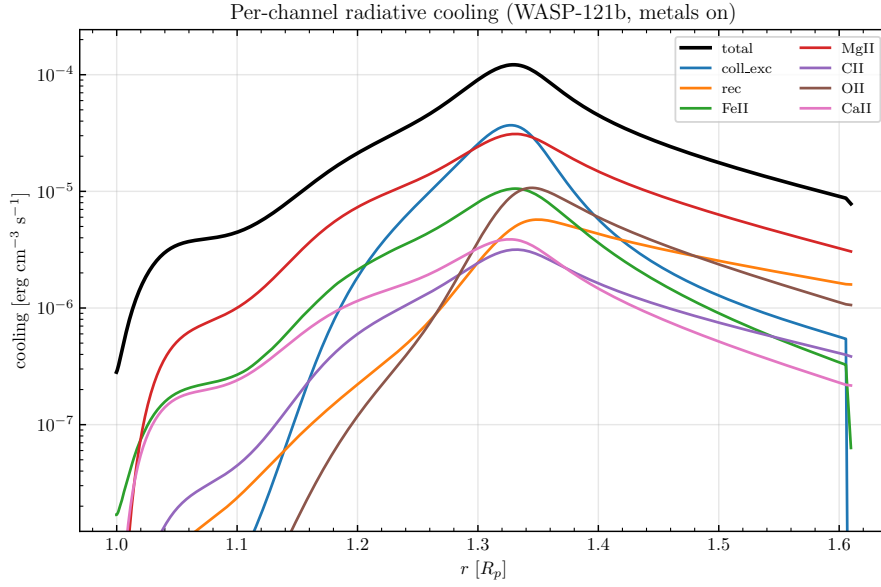


Figure 5: Radiative cooling by channel vs. radius (WASP-121b, metals on; from `Cooling_breakdown.txt`). Below $\sim 1.4 R_p$ the metal lines (e.g. Fe II, Mg II, C/O) compete with or dominate the H/He collisional channels.

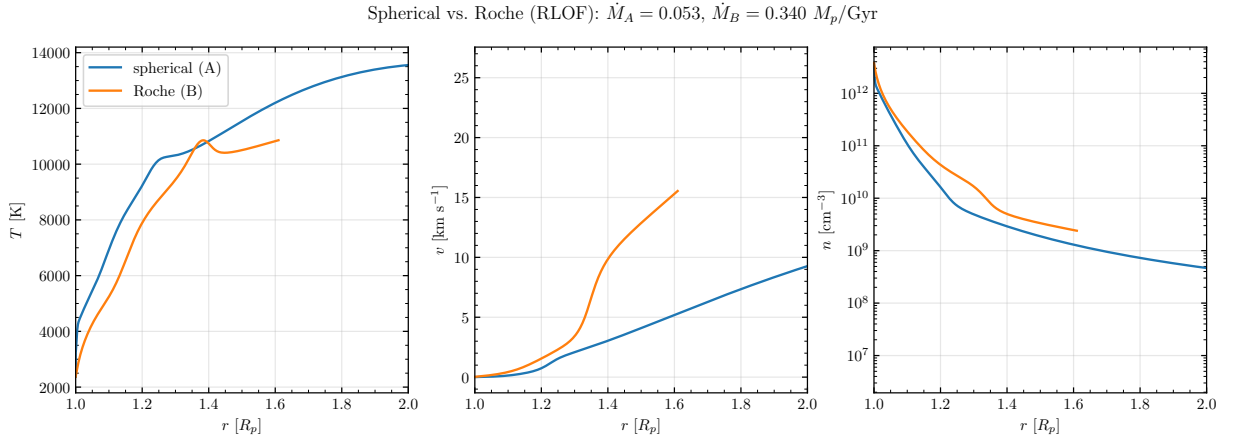


Figure 6: Spherical (Case A) vs. Roche (Case B) WASP-121b: temperature, velocity, and density. The Roche wind reaches L1 ($\approx 1.6 R_p$) and accelerates much faster; $\dot{M}$ is annotated. The temperature panel is the `_adv` profile; its base is smooth because the post-processor falls back to the equilibrium solution where the flow is not a clean outflow (see the caveat in §4.). The Roche velocity (middle) dips slightly negative near the base — the genuine, physical “breathing” inflow of a near-Roche-filling planet.

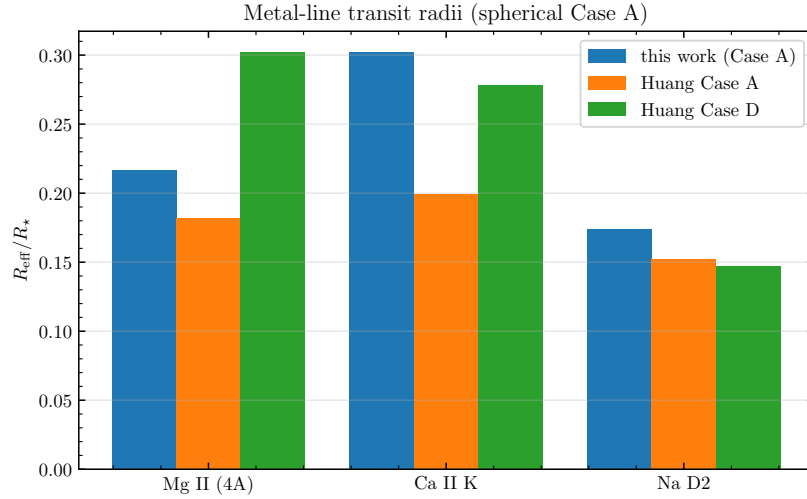


Figure 7: Metal-line transit radii R_{eff}/R_* for spherical Case A, compared with Huang et al. (2023) Case A and Case D.

8. Atomic and molecular data

By default EXHALE is an *atomic/ionic* code: the modeled species are H, He (including the He 2^3S metastable triplet), and ten trace metals (C, N, O, Mg, Si, Ca, Na, K, S, Fe) across their first two–three ionization stages. The optional molecular network (Molecular chemistry: True, §3.7) adds H_2 , H_2^+ , H_3^+ and HeH^+ to the coupled equilibrium, with H_2 photoionization opacity (Yan et al. 1998) and H_3^+ infrared cooling (Miller et al. 2013). Beyond those four there is **no molecular chemistry and no molecular opacity** — no H^- , CO, or H_2O — so no molecular line lists or partition functions enter the calculation. Table 7 collects the provenance of every rate and cross section, grouped by physical process. The H/He baseline is inherited essentially unchanged from the original ATES (Caldirola et al. 2021, 2022); the metal, excited-hydrogen, and charge-exchange data were added in this work.

Table 7: Atomic data sources, by process. Source files: `cross_sec.f90` (cross sections); `Cool_coeff.f90` (collisional rates, recombination, cooling); `charge_exchange.f90`; `excited_hydrogen.f90`; `EXHALE_transit.py` (transmission line data).

Process	Species	Source
Elemental abundances		
Solar reference	10 metals (metals.inp template)	Asplund et al. (2009)
Helium	He/H ratio	input parameter (He/H number ratio)
Photoionization cross sections		
Bound–free $\sigma(E)$	HI	hydrogenic (Kramers) analytic form (orig. ATES)

Table 7, continued

Process	Species	Source
Opacity model (runtime)	He I	Verner, Ferland, Korista & Yakovlev (1996) default; legacy ATES 2-term fit via ATES_photoionization_rate
	He I 2^3S (metastable)	two VFKY96 (Verner) wings + power-law bridge, fitted to TOPbase/Opacity-Project (extends the orig. Norcross 1971 broken power law past 60 eV)
	C, N, O, Mg, Si, Na, Fe (+ ions)	Verner, Ferland, Korista & Yakovlev (1996), single-shell fits
	K I	Verner & Yakovlev (1995)
	all (via opacity.inp)	A=analytic above (default); C=constant above threshold, $\sigma = \text{OPA_CONST_X}$ times the analytic value at the edge; P=that constant base times the pressure-broadening factor (Robinson & Catling); T=tabulated files
Collisional (electron-impact) ionization		
Rate $C(T)$	H I, He I, He II	Voronov (1997, ADNDT 65, 1) default; legacy ATES fits (Abel+1997 / Hui & Gnedin 1997) via Legacy_HHe_rates
	C–Fe, all stages	Voronov (1997, ADNDT 65, 1)
Recombination (radiative RR + dielectronic DR)		
Case B (RR)	H II, He II, He III	Badnell (2023) RR – Mao & Kaastra (2016) α_1 (+ Badnell DR for He II) default; legacy Hui & Gnedin (1997) fits via Legacy_HHe_rates
He rec. radiation $\rightarrow$ H I (def. on, He_rec_coupling)	He II $\rightarrow$ He I	Draine (2011) on-the-spot y/z: extra H I photoionization $n_{\text{HeII}}n_e[z\alpha_B + y\alpha_1]$ with photoelectron heating, He II recombination shifted to $\alpha_B + y\alpha_1$; False = pure case B
Metastable	He II $\rightarrow 2^3S, 1^1S$	power-law fits (orig. ATES)
Penning ionization	He $2^3S + H$	Taylor et al. (2025) temperature-dependent fit (replaces the const. 5×10^{-10} ; always on)

Table 7, continued

Process	Species	Source
	He 2 ³ S + H ₂	fit to García Muñoz (2025) Table A.5 (penning_HeI23S_H2, valid 500–10 000 K): He 2 ³ S + H ₂ → He 1 ¹ S + H ₂ ⁺ + e [−] , releasing 4.4 eV to the electrons. The minor H + HeH ⁺ associative-ionization branch is not resolved — the full rate goes to the dominant Penning channel. Active only when a molecular run also tracks the triplet
RR + DR	C, N, O, Mg, Si, Na, K, S, Ca II (+ stages)	Badnell (2006, ApJS 167, 334) RR + Badnell adf48 DR (TAMOC clist_K)
RR only	Ca I (Ca II+e)	Shull & Van Steenberg (1982); no DR
RR + DR	Fe I, Fe II	Huang et al. (2023) analytic fit; the Badnell DR project does not cover these isoelectronic sequences (see the note below the table)
Charge exchange (charge transfer)		
metal + H/H ⁺ (default)	23 reactions (Group A)	Huang et al. (2023, ApJ 951, 123) Table 4
full network (cx_full)	+ He+H, metal+He, metal+metal	Huang et al. (2023) Table 4 (Groups B/C/D)
Radiative cooling		
recomb., coll. ion./exc., free–free	H, He	Hui & Gnedin (1997) recomb. cooling; Black (1981) collisional; free–free bremsstrahlung with the van Hoof et al. (2014) thermal Gaunt factor, weighted by ion net charge
H ₃ ⁺ infrared (molecular runs)	H ₃ ⁺	Miller et al. (2013) LTE emission per molecule (Table 5 $z(T)$ fits) times their Table 6 non-LTE departure factor $s(T, n_{\text{H}_2})$; optically thin. The dominant coolant of a molecular base, where every atomic channel above is exponentially suppressed
metal line cooling (analytic)	C I/II, N I/II, O I/II	closed-form fits to CHIANTI v11.0.2 (default); legacy AIOLOS fits via cno_cool 0 (no N cooling)
metal line cooling (analytic)	Mg I/II, Ca II H&K, Na I D, Fe II coronal	closed-form fits to CHIANTI v11.0.2 (Burgess–Tully descaled; 1–3% accuracy); Na I D via effective $\bar{g} \approx 0.2$
metal line cooling (tabulated)	Fe I; Fe II (T, n_e)	NIST f -values + Van Regemorter; CHIANTI multilevel statistical equilibrium

Table 7, continued

Process	Species	Source
ground-term fine structure (statistical equilibrium)	C I $2p^{2^3}P$, C II $2p^2P$, N II $2p^{2^3}P$, O I $2p^{4^3}P$	the split ground term is solved in exact statistical equilibrium at the local (n_e, n_{HI}) instead of the low-density coronal limit, which overstates these lines by 2.9–8.3 decades at the base. Electron collision strengths from CHIANTI v11.0.2; H-atom rates from Yan & Babb (2023) for C I/N II, Abrahamsson, Krems & Dalgarno (2007) for O I, Barinovs et al. (2005) for C II. The coronal curve is kept only for the channels leaving the ground term. The complete set: the other coolants have a single ground level, are built from permitted lines (Fe I), or already use a statistical-equilibrium table (Fe II)
line trapping	the eight ground-term fine-structure lines	line-center escape probability from the column above the cell, Hollenbach & McKee (1979) / de Jong, Boland & Dalgarno (1980) plane-parallel Doppler form renormalized to $\beta(0) = 1$; applied as $A_{ul} \rightarrow \beta A_{ul}$ inside the statistical-equilibrium solution, the only placement correct in both the low- and high-density limits. Other lines, including Mg II h&k, are treated as optically thin
validity floor of the coronal fits	all CHIANTI-derived coefficients	the coronal part is cut off smoothly below 10^3 K (the lower end of the fitted range), continuous in value and dT -slope and bit-identical above it; only the explicitly solved ground-term fine-structure terms survive below. The roll-off is $\exp(-x^2)$ with $x = (T_{\text{floor}}/T - 1)/w$ and $w = 0.1$ by default, settable as <code>Coronal cutoff width:</code> . With the C/N/O ground-term fine-structure floors saturated no coronal cooling survives below the floor for the guard to remove, and the base-cell balance temperature of all four paper planets is identical over $w = 0.02$ – 1.2 . Not applied to the legacy AIOLOS branch (<code>cno_cool 0</code>)
Photoelectric heating		

Table 7, continued

Process	Species	Source
secondary ionization (def. on, Secondary_ionization)	HI, HeI	Shull & van Steenberg (1985) $f_{\text{heat}}(x)$, $f_{\text{ion}}(x)$ fits; photoelectrons with $E_0 > 40\text{eV}$ drive secondary HI / HeI ionizations, heating scaled to f_{heat} ; the He 2^3S photoelectron feeds the same heating and secondary-source budget
Excited hydrogen H($n=2$) and Lyα		
$2s/2p$ rate equilibrium	H($n=2$)	Christie, Arras & Li (2013) Table 2
recombination cascade to $n=2$	H	Draine (2011)
Ly α pumping field (mode 0)	H	Huang et al. (2017) Eq. 6
Transmission line data (EXHALE_transit.py)		
λ , f -value, A_{ki}	He 10830, Ly α , H α , H β , Mg II, Ca II H&K, Na I D	NIST ASD
line profile	all	Voigt (Faddeeva, scipy.special.wofz)

Note on Fe. Iron recombination uses the Huang et al. (2023) analytic RR+DR fit while every other metal uses the Badnell tabulation, and this is not a deferred item: no Badnell fit exists for either stage. Fe I is formed by recombining Fe II (Mn-like, 25 electrons) and Fe II by Fe III (Cr-like, 24), whereas the Badnell dielectronic-recombination project reaches the phosphorus isoelectronic sequence (15 electrons) as of its paper XVI (2022). Ca I has the same problem one sequence lower (K-like, 19 electrons) and likewise keeps a non-Badnell rate. **Note on completeness.** The metal grid carries the stages needed for the XUV-driven ionization balance and the observed NUV/optical lines; higher stages and additional elements (e.g. the full Fe forest, K I, Mg I) beyond those tabulated above are not modeled.